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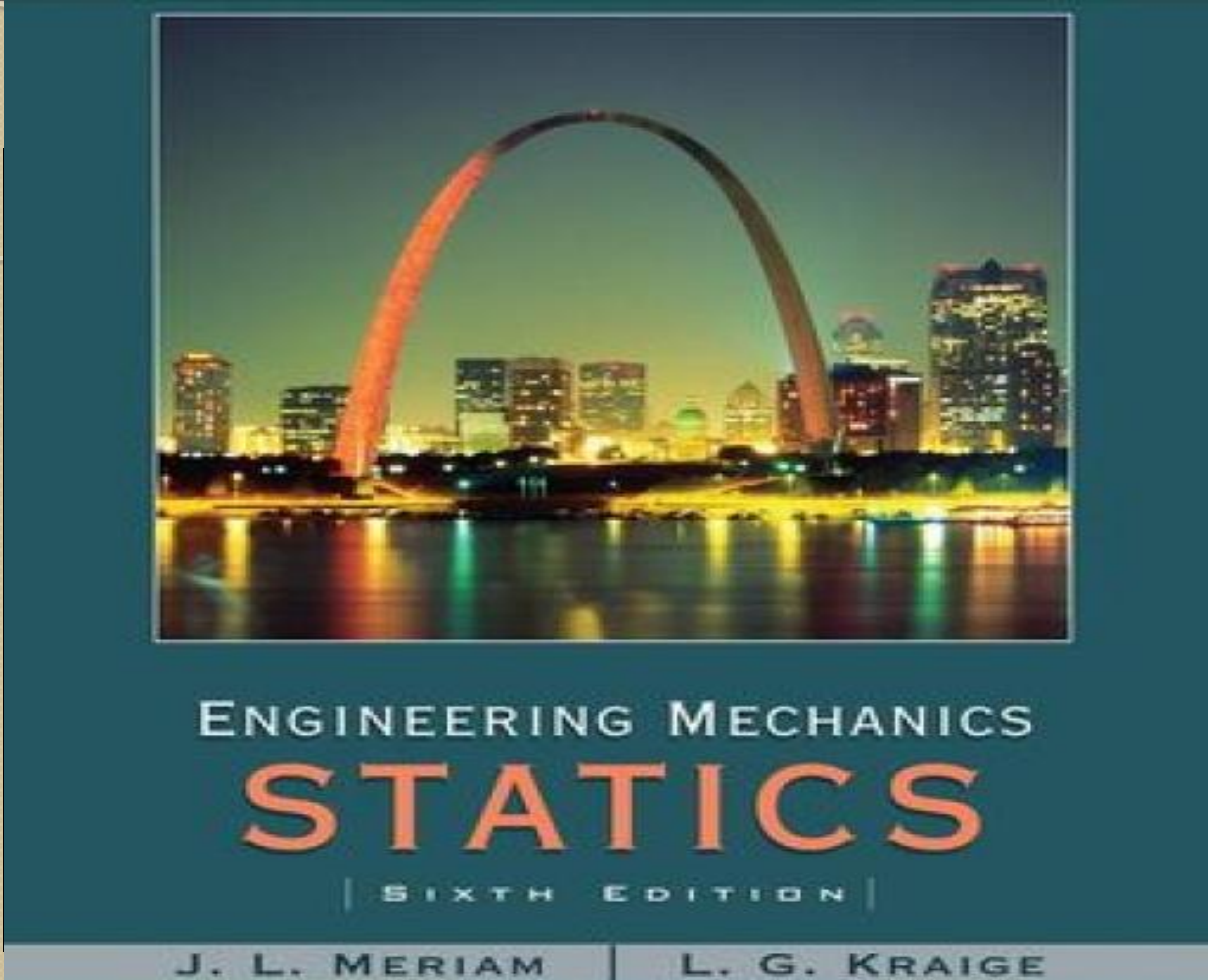

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2 FORCE SYSTEMS

2. Force Systems

Introduction

a. Force Systems

2-D Force Systems

a. Rectangular Components

b. Moment

c. Couple

d. Resultants

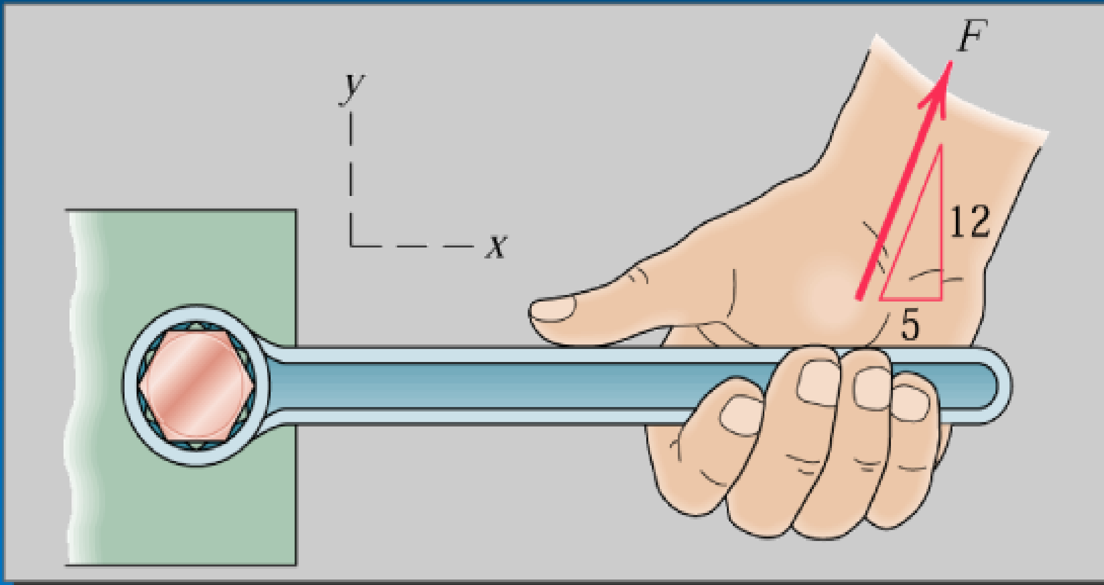
3-D Force Systems

a. Rectangular Components

b. Moment

c. Couple

d. Resultants



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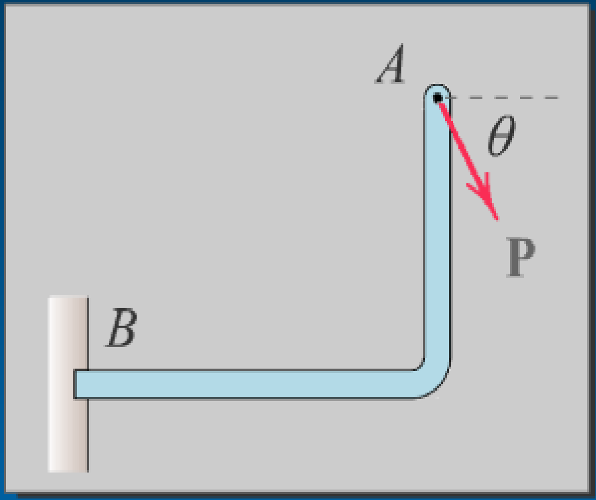
Force Systems

Force: Action of one body on another - a push or pull.

Force is a vector quantity, characterized by

- magnitude (P)
- direction (θ)
- point of application (A)



Changing any one of the above items would, in general, change the effects of the force **P** as "felt by" the support at B.

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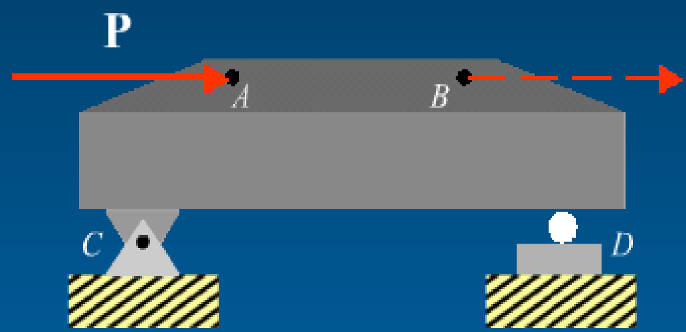
Principle of transmissibility: A force may be moved along its line of action without changing the external reaction on the body. (Internal effects may indeed change, however).



Force **P** may be moved from *A* to *B* without changing external reactions acting at *C* and *D*.

Click on solid arrow to move force **P**

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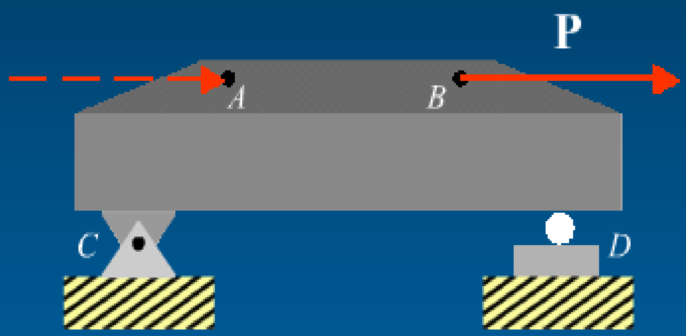
Principle of transmissibility: A force may be moved along its line of action without changing the external reaction on the body. (Internal effects may indeed change, however).



Force **P** may be moved from *A* to *B* without changing external reactions acting at *C* and *D*.

Click on solid arrow to move force **P**

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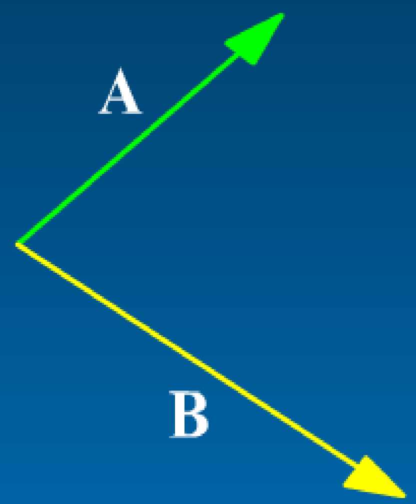
Forces are vector quantities; therefore, adding concurrent forces follows the rules for vector addition.



Parallelogram Rule

Triangle Rule

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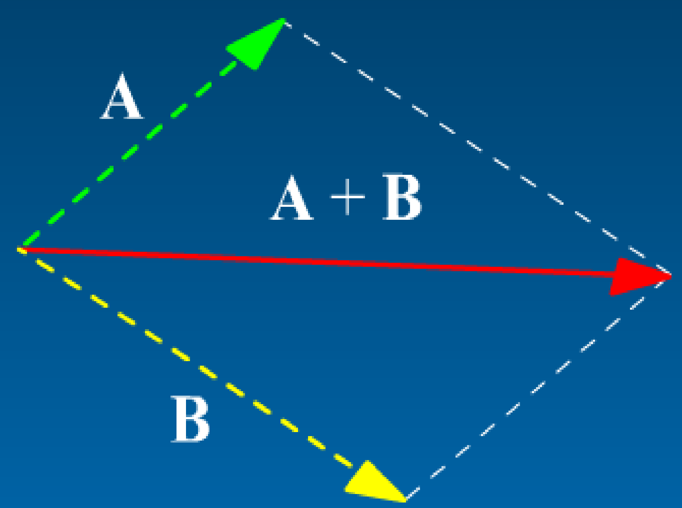
Forces are vector quantities; therefore, adding concurrent forces follows the rules for vector addition.



<More>
Triangle Rule

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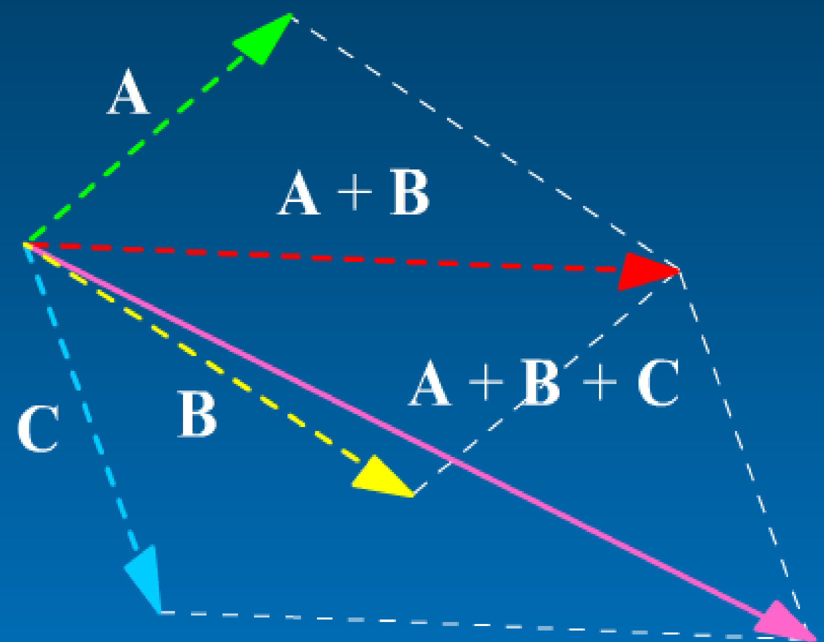

Parallelogram Rule

Triangle Rule

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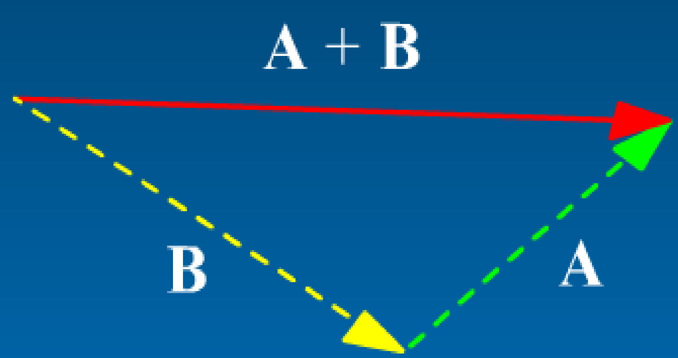

Parallelogram Rule

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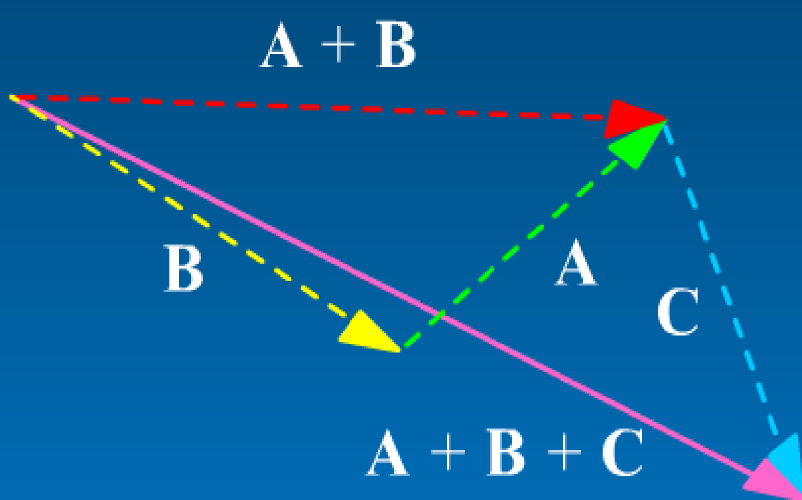
Forces are vector quantities; therefore, adding concurrent forces follows the rules for vector addition.

Parallelogram Rule

Triangle Rule



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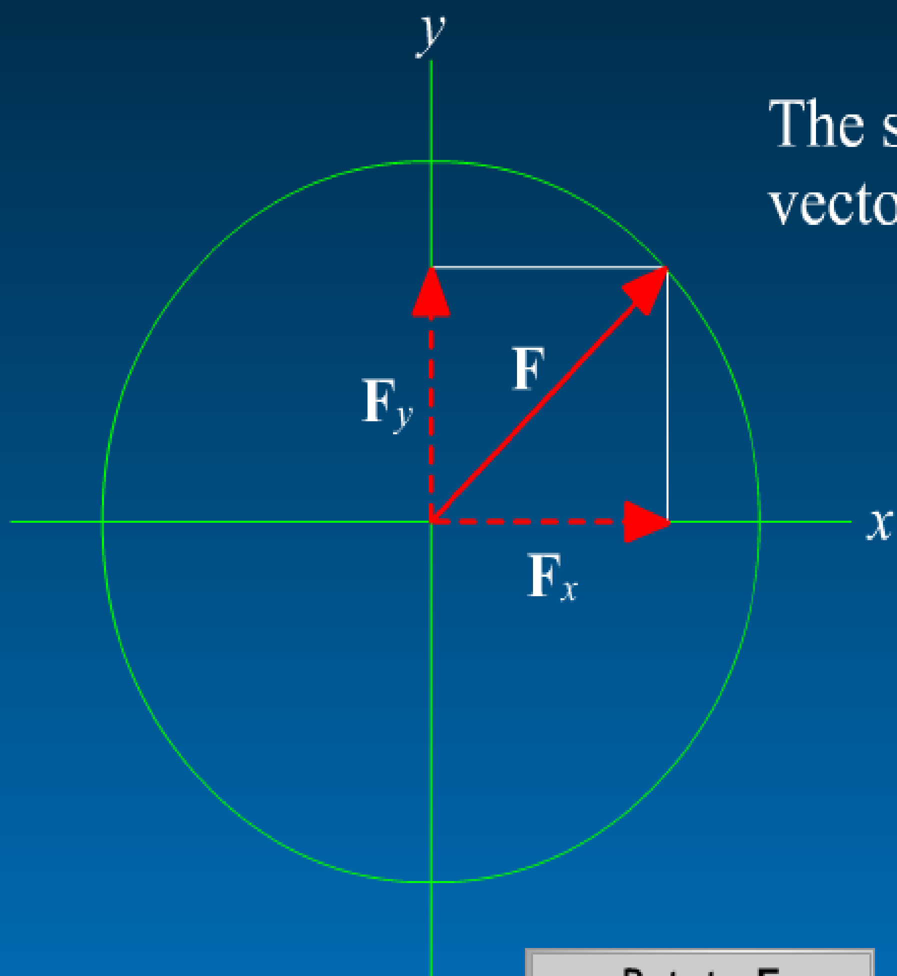

The scalar components of a vector include sign information:

F_x	F_y
+	+

Rotate F

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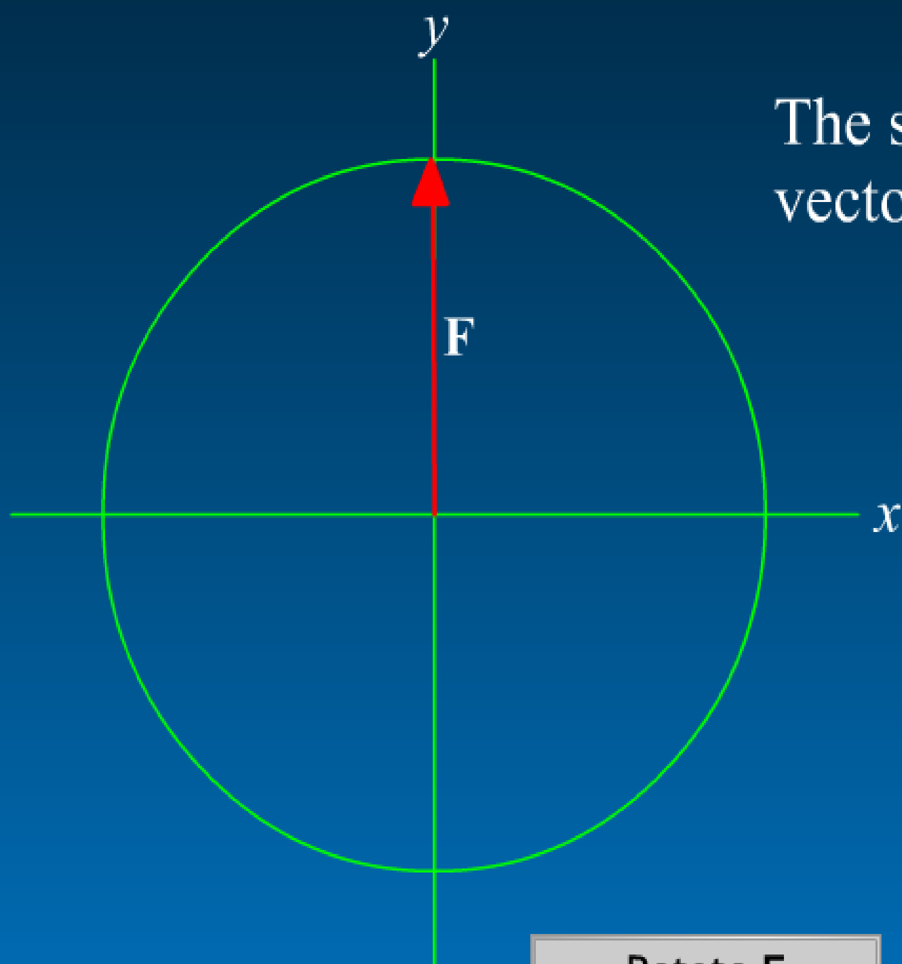
The scalar components of a vector include sign information:



F_x	F_y
	+

Rotate F

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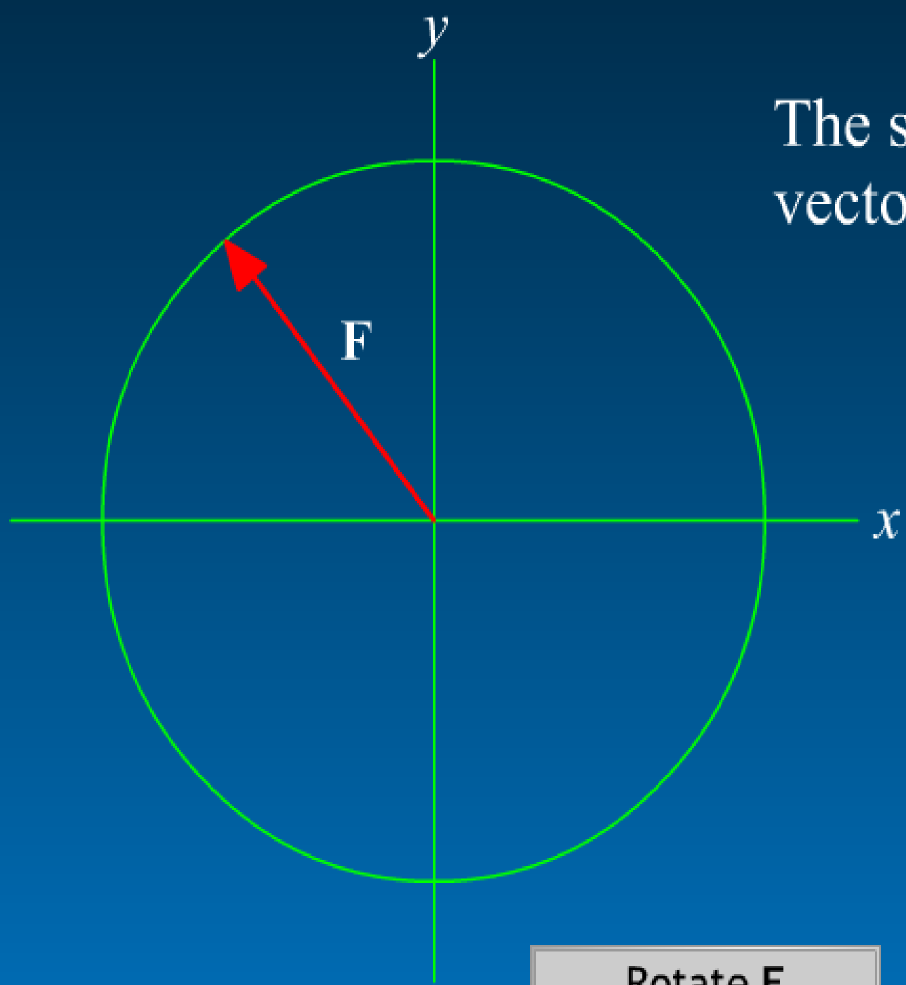

The scalar components of a vector include sign information:

F_x	F_y
−	+

Rotate F

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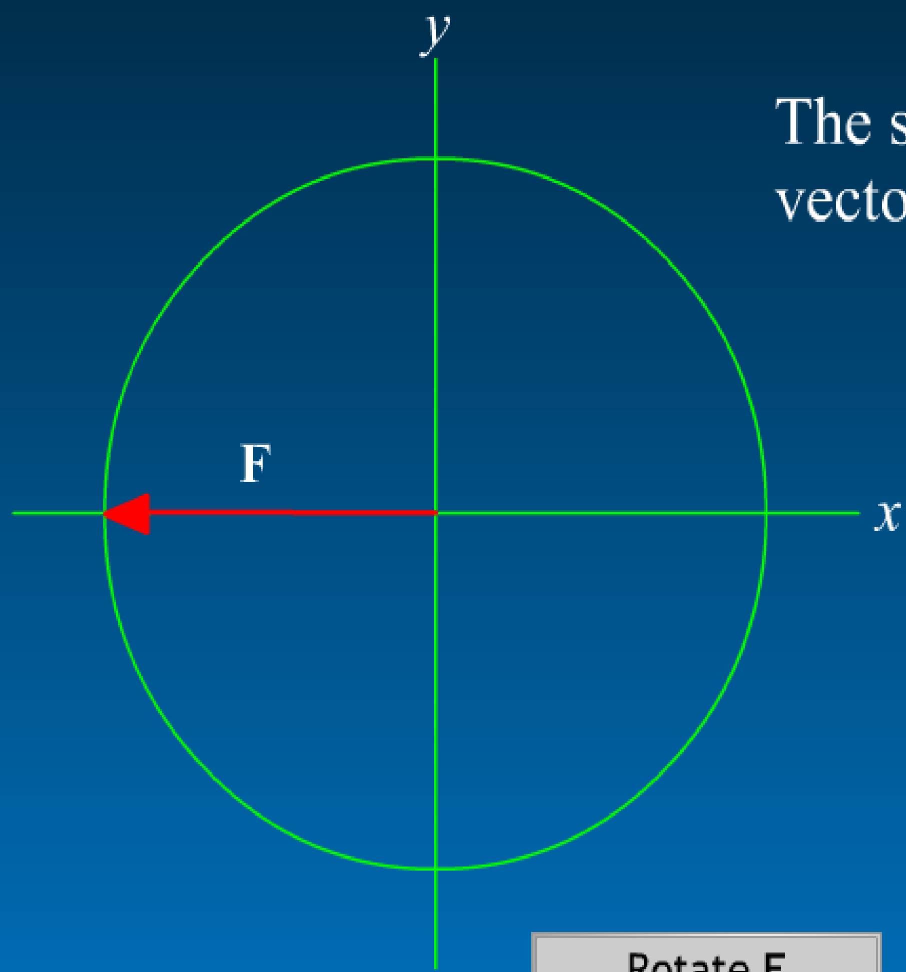

The scalar components of a vector include sign information:

F_x	F_y
—	

Rotate F

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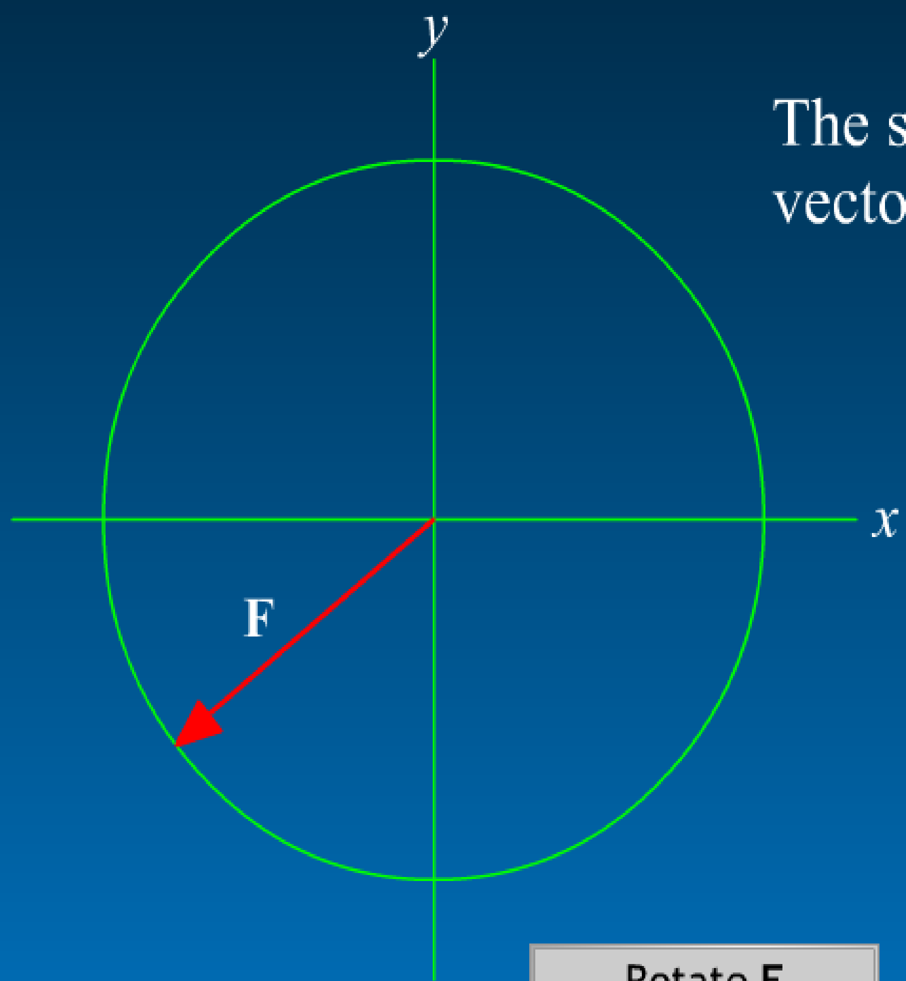

The scalar components of a vector include sign information:

F_x	F_y
—	—

Rotate F

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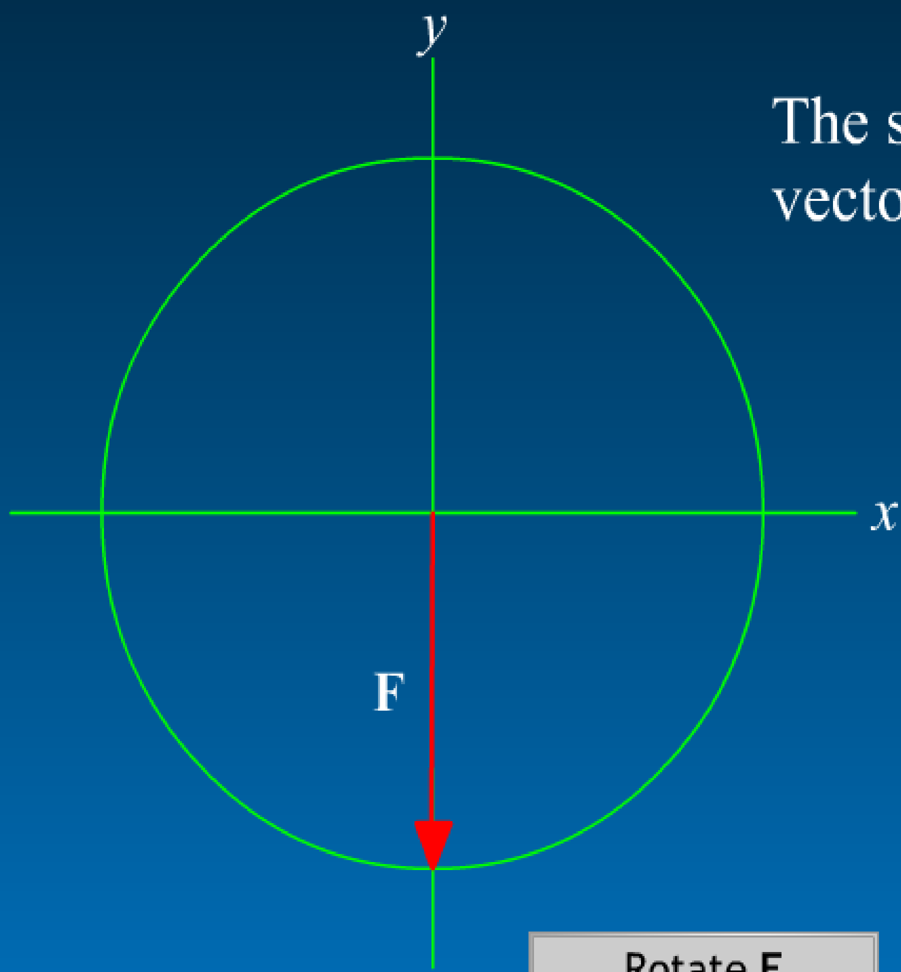
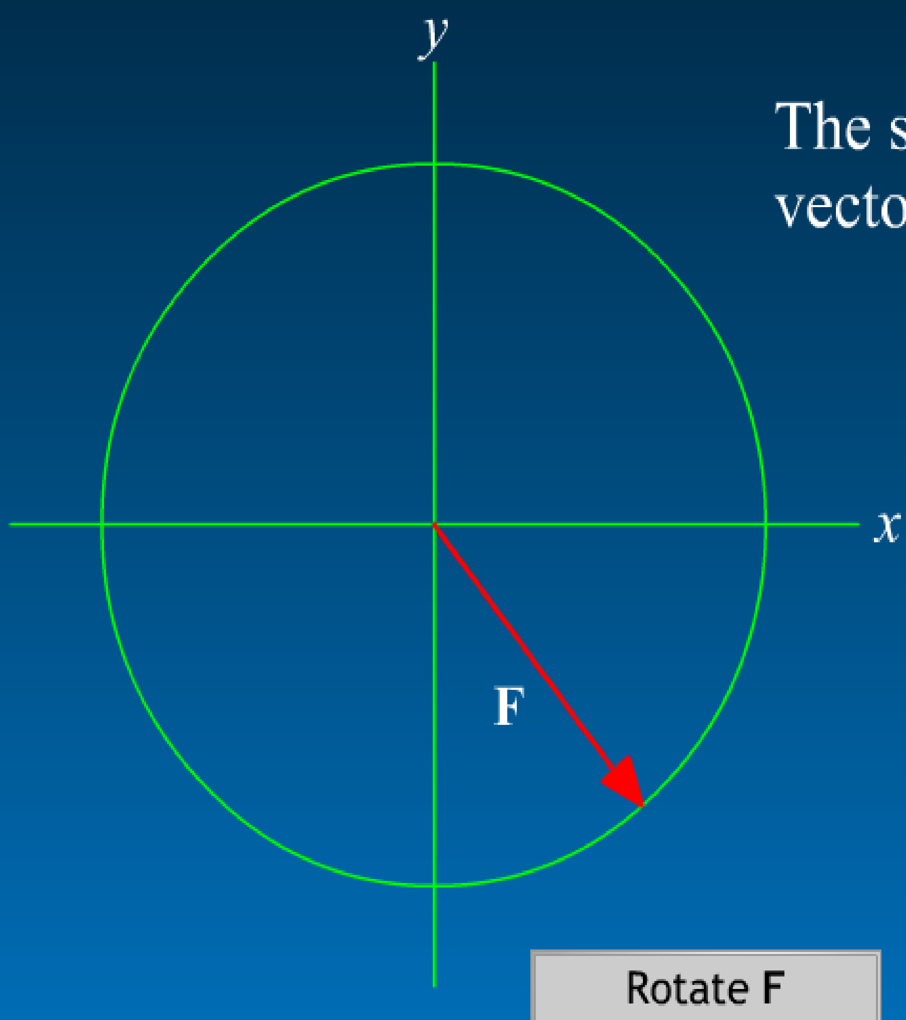


The scalar components of a vector include sign information:

F_x	F_y
	–

Rotate F



The scalar components of a vector include sign information:

F_x	F_y
+	−

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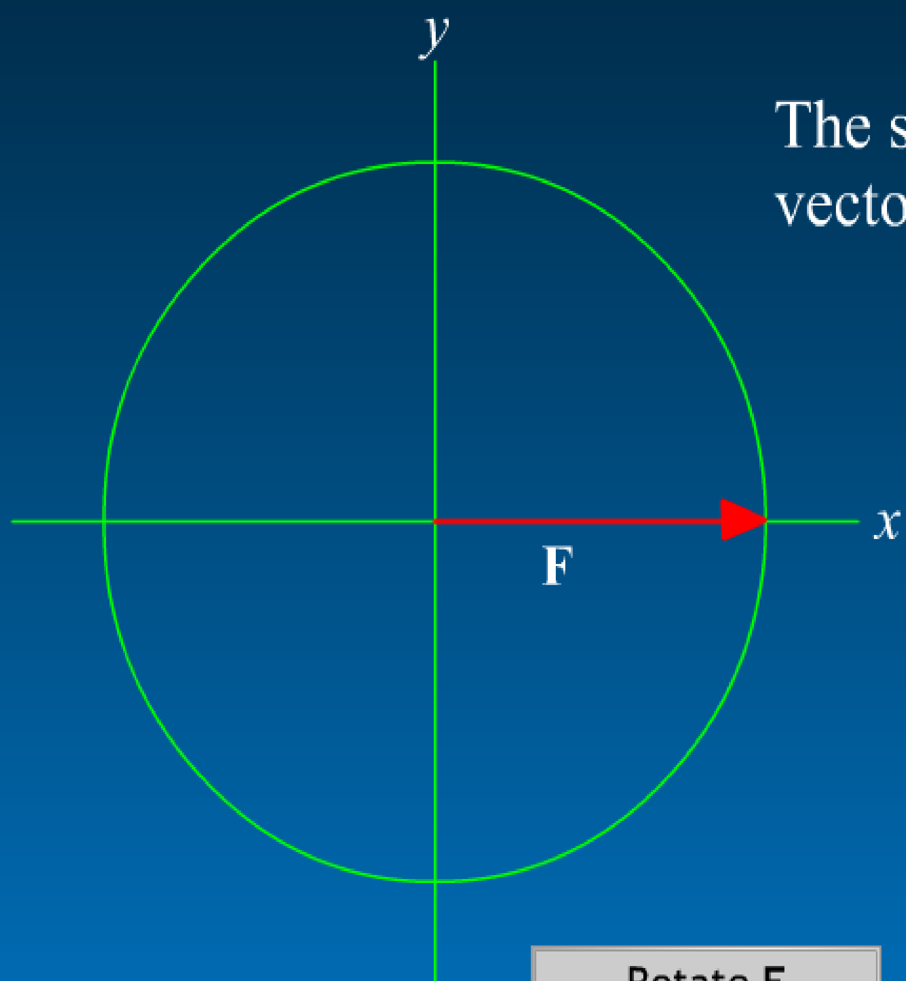

The scalar components of a vector include sign information:

F_x	F_y
+	

Rotate F

Addition of Vectors by Summation of Scalar Components

$$\mathbf{R} = R_x\,\mathbf{i} + R_y\,\mathbf{j}$$

$$\Sigma\mathbf{F} = \sum_{i=1}^{n}(F_{i,x}\,\mathbf{i} + F_{i,y}\,\mathbf{j}) = (\Sigma F_x)\,\mathbf{i} + (\Sigma F_y)\,\mathbf{j}$$

$$\therefore\; R_x = \Sigma F_x$$

$$R_y = \Sigma F_y$$

A term such as ΣF_x should be thought as "the algebraic sum of the x scalar components."

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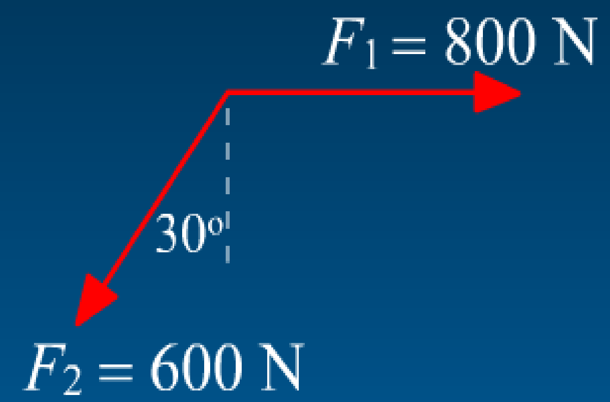
Example: Add the two given forces.

I. Parallelogram rule

II. Rectangular components



Click on yellow headings for solutions

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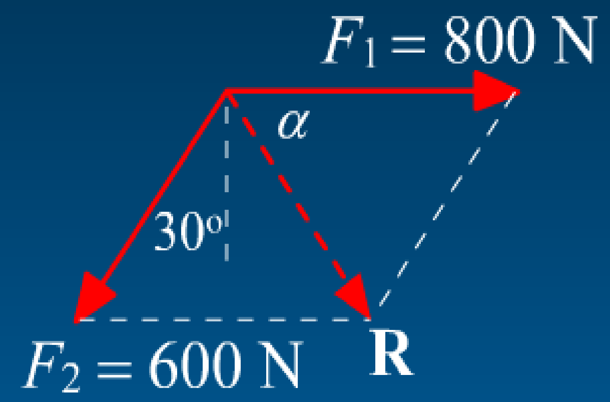
Example: Add the two given forces.

I. Parallelogram rule



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Example: Add the two given forces.

I. Parallelogram rule

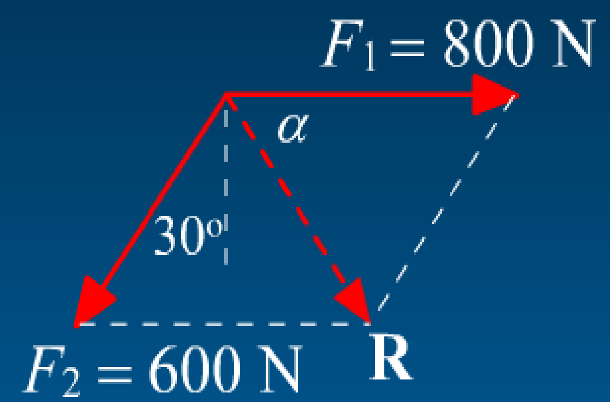
By the use of the law of cosines

$$R^2 = 800^2 + 600^2 - 2\,(800)\,(600)\cos 60^{o}$$

$$R = 721 \text{ N}$$

By the use of the law of sines

$$\frac{\sin\alpha}{600} = \frac{\sin 60^{o}}{721}\ ,\ \alpha = 46.1^{o}$$



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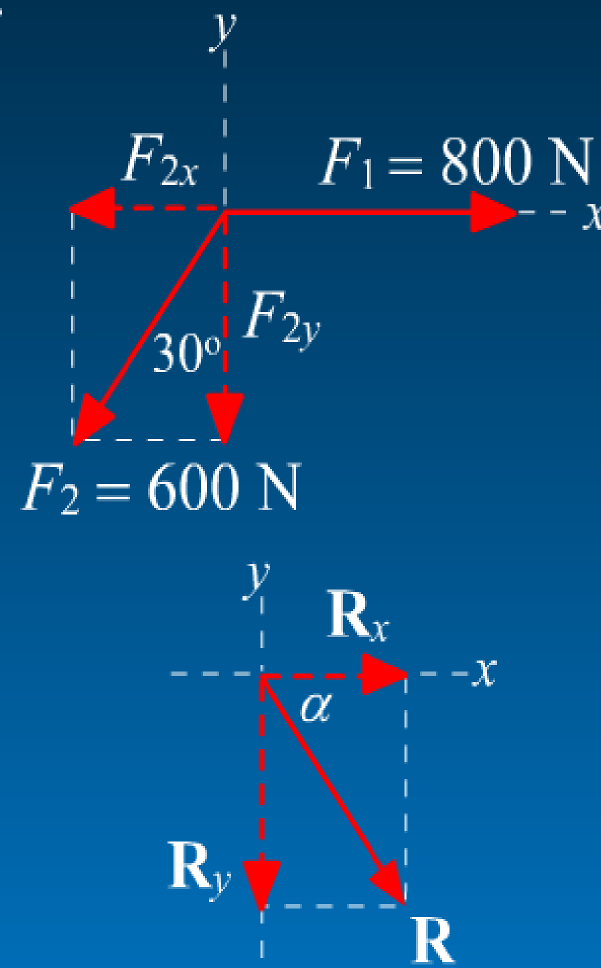
Example: Add the two given forces.

II. Algebraic Solution



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Example: Add the two given forces.

II. Algebraic Solution

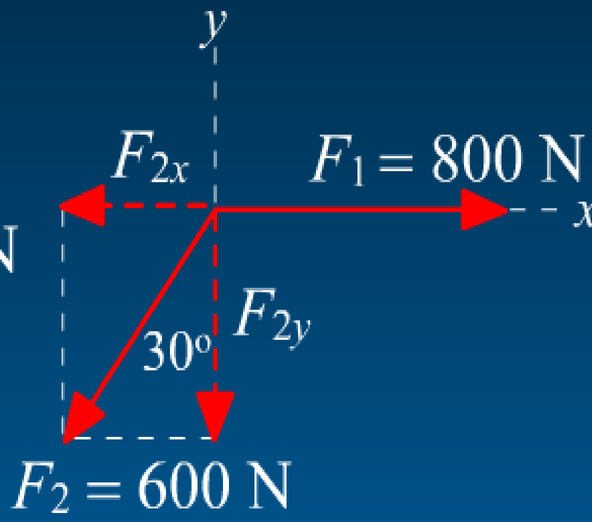
$$R_x = \Sigma F_x = 800 - 600 \sin 30^\circ = 500 \text{ N}$$

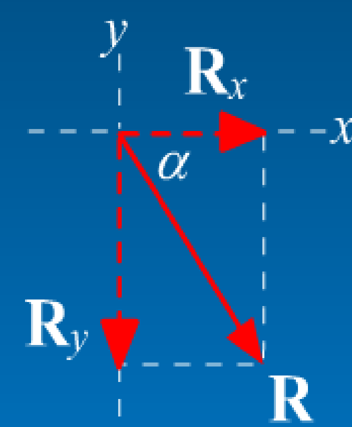
$$R_y = \Sigma F_y = 0 - 600 \cos 30^\circ = -520 \text{ N}$$

$$\mathbf{R} = R_x \mathbf{i} + R_y \mathbf{j} = (500\,\mathbf{i} - 520\,\mathbf{j}) \text{ N}$$

$$R = \sqrt{R_x^2 + R_y^2} = 721 \text{ N}$$

$$\alpha = \tan^{-1} \frac{|R_y|}{|R_x|} = 46.1^\circ$$





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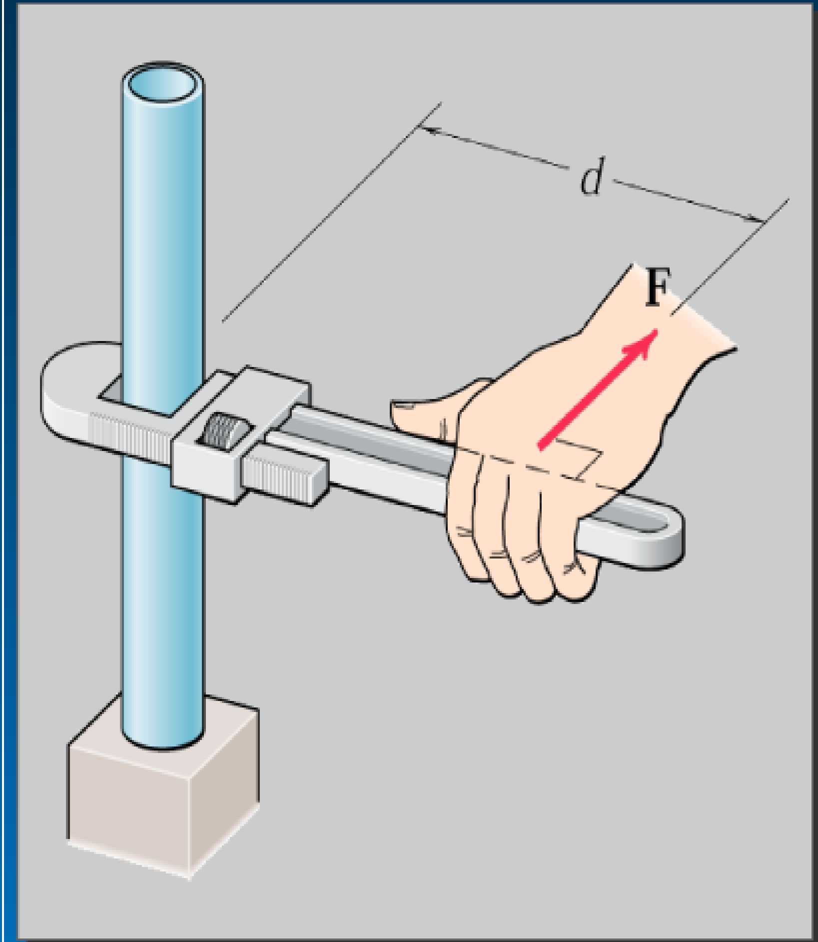
Moment



A moment is the tendency of a force to rotate a body about some axis.

In this example, the turning effect of force **F** clearly depends on:

* the magnitude of **F**
* the length of the wrench handle

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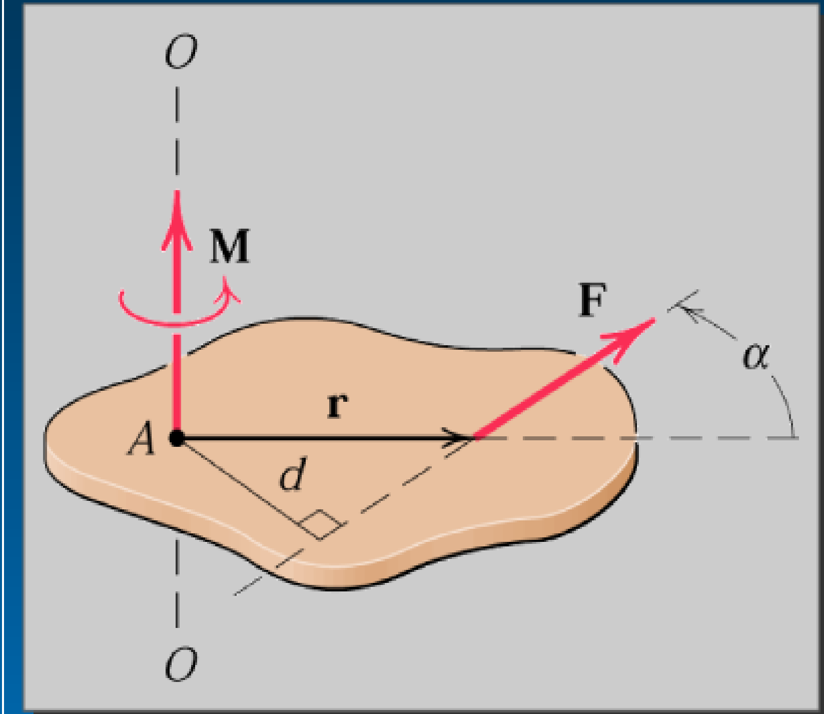

The moment of **F** about point *A* is formally defined as the vector

$$\mathbf{M}_A = \mathbf{r} \times \mathbf{F}$$

where **r** is a position vector which runs from point *A* to any point on the line of action of **F**.

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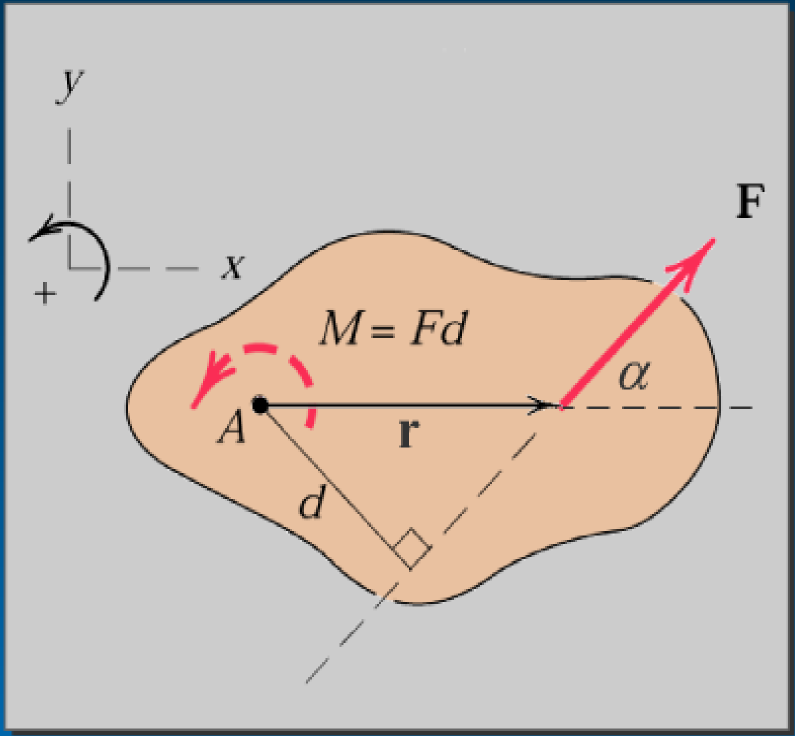
2 - D View:



$$\mathbf{M}_A = \mathbf{r} \times \mathbf{F}$$

The magnitude of $\mathbf{M}_A$ is

$$M_A = rF\sin\alpha = F(r\sin\alpha)$$
$$= Fd$$

where d is the perpendicular distance or *moment arm* from A to the line of action of **F**.

Example: Determine the magnitude of the moment about O of the 600-N force in five different ways.

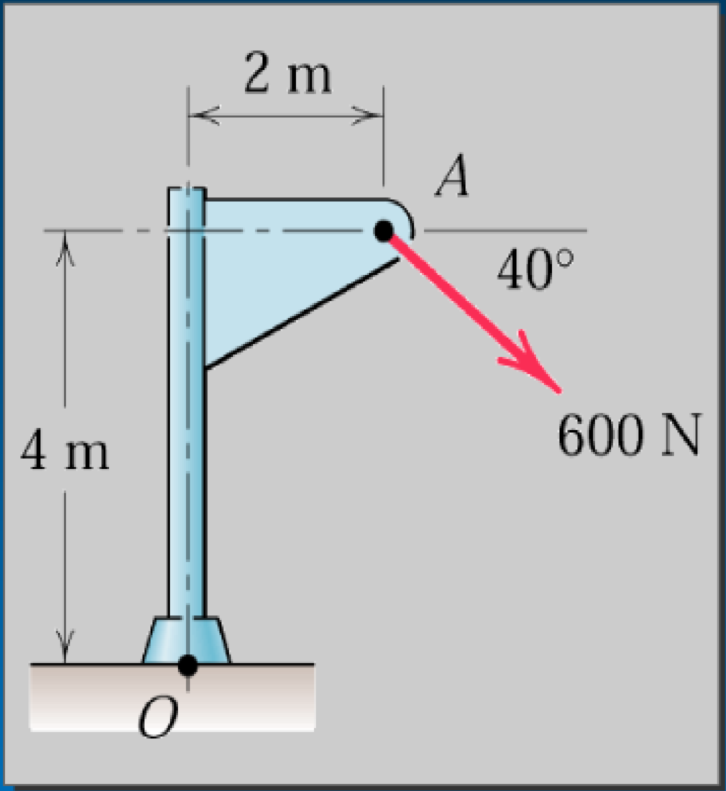
I. $\mathbf{M}_O = \mathbf{r} \times \mathbf{F}$

II. $M = Fd$

III. Varignon's Theorem

IV. Move force to ground level

V. Move force to above O



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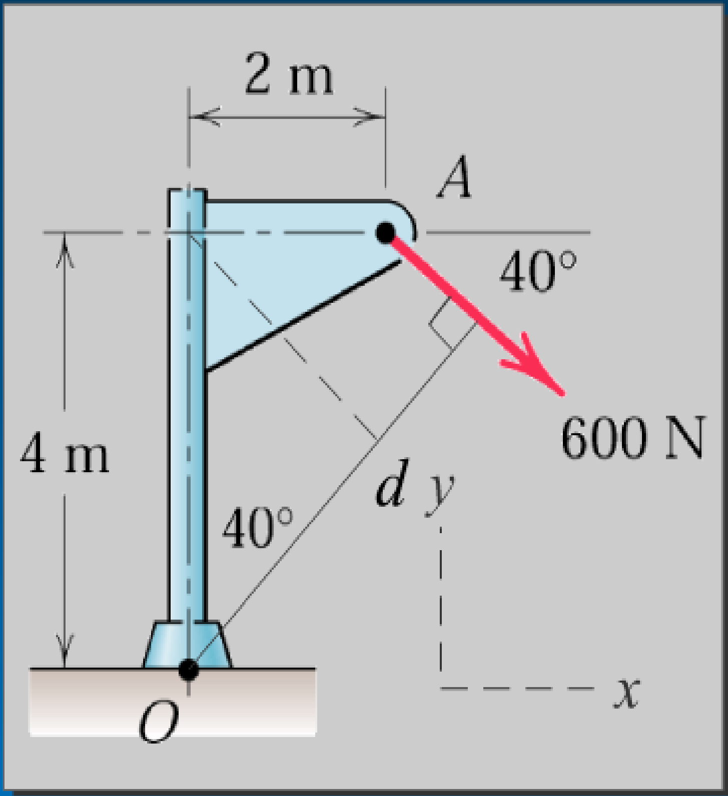
Example: Determine the magnitude of the moment about O of the 600-N force in five different ways.

II. $M = Fd$

$d = 4\cos 40^{\circ} + 2\sin 40^{\circ}$

$= 4.35\ \text{m}$

$+ \; M = 600\,(4.35) = 2610\ \text{N} \cdot \text{m}$



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Example: Determine the magnitude of the moment about O of the 600-N force in five different ways.

III. Varignon's Theorem

The rectangular components of the 600-N force are

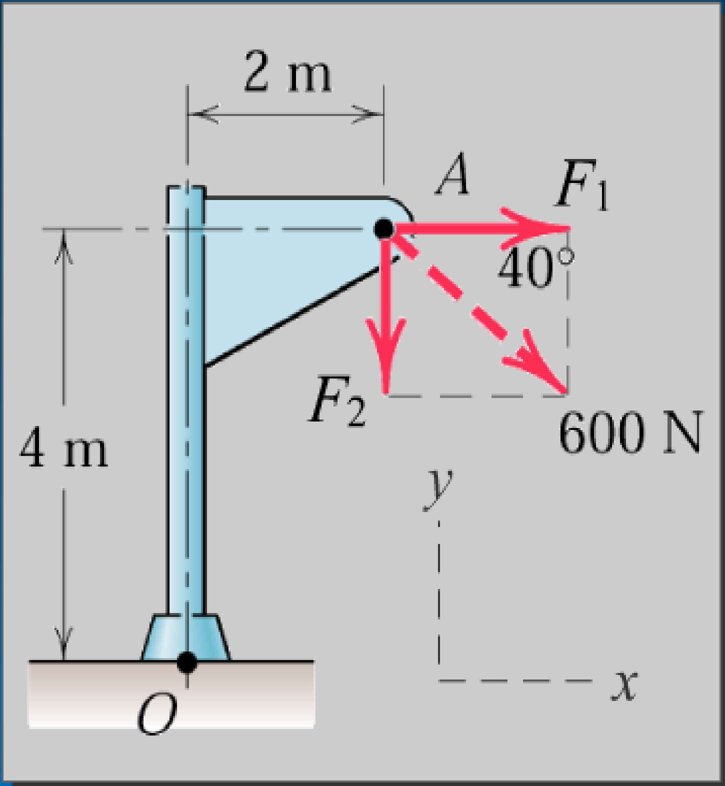
$$F_1 = 600\cos 40^\circ = 460\ \text{N}$$

$$F_2 = 600\sin 40^\circ = 386\ \text{N}$$

Then

$$+M = 460\,(4) + 386\,(2)$$

$$= 2610\ \text{N}\cdot\text{m}$$



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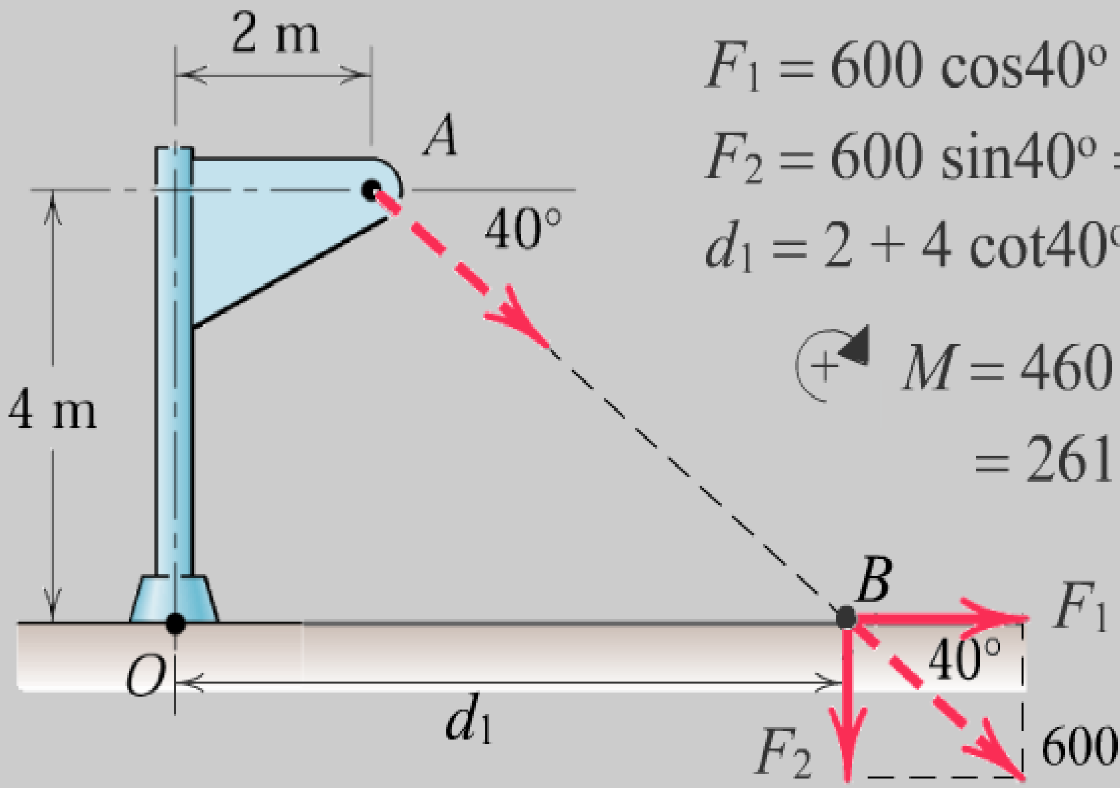
Example: Determine the magnitude of the moment about O of the 600-N force in five different ways.

IV. Move force to ground level



$$F_1 = 600\cos 40^\circ = 460 \text{ N}$$

$$F_2 = 600\sin 40^\circ = 386 \text{ N}$$

$$d_1 = 2 + 4\cot 40^\circ = 6.77 \text{ m}$$

$$(+)\; M = 460\,(0) + 386\,(6.77)$$

$$= 2610 \text{ N}\cdot\text{m}$$

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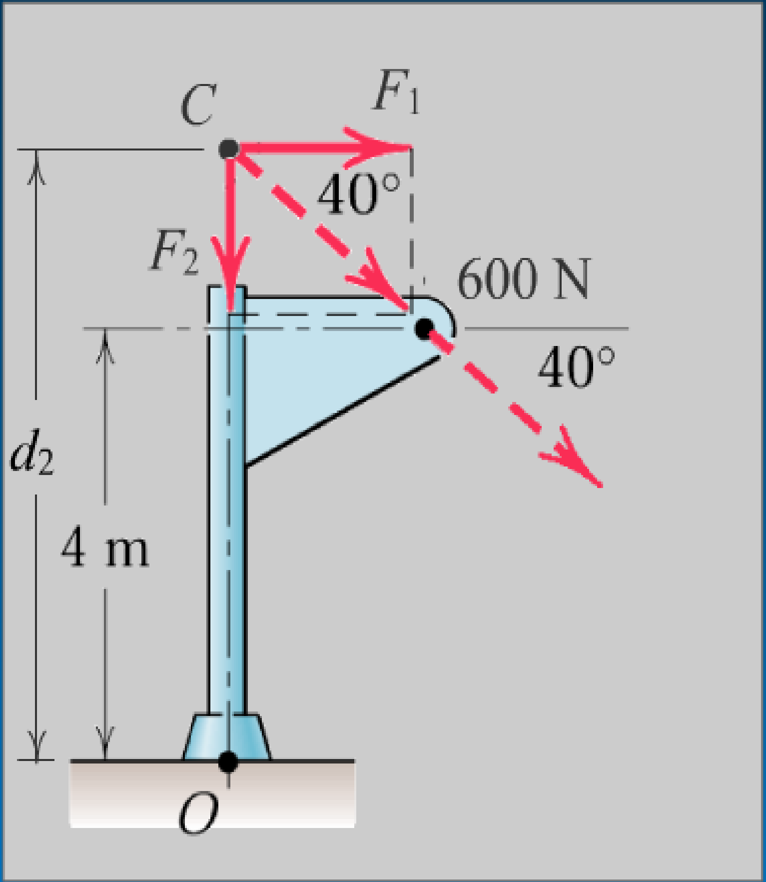
Example: Determine the magnitude of the moment about O of the 600-N force in five different ways.

V. Move force to above O

$$F_1 = 600 \cos 40^\circ = 460 \text{ N}$$
$$F_2 = 600 \sin 40^\circ = 386 \text{ N}$$
$$d_2 = 4 + 2 \tan 40^\circ = 5.68 \text{ m}$$

$$M = 460\,(5.68) + 386\,(0)$$
$$= 2610 \text{ N} \cdot \text{m}$$



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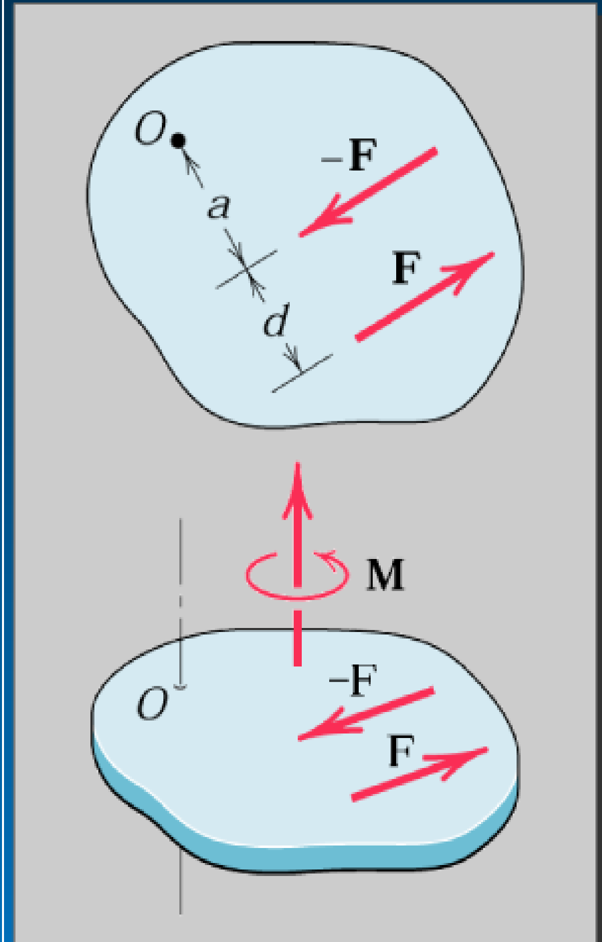
Couple



A *couple* consists of two equal and opposite forces **F** and **-F** separated by a non-zero perpendicular distance *d*.

The moment associated with the couple shown in the upper figure is,

$$M_O = F(a+d) - Fa$$
$$= Fd$$

The moment is independent of the location of the moment center *O*. Therefore the moment associated with a couple is a *free vector.*

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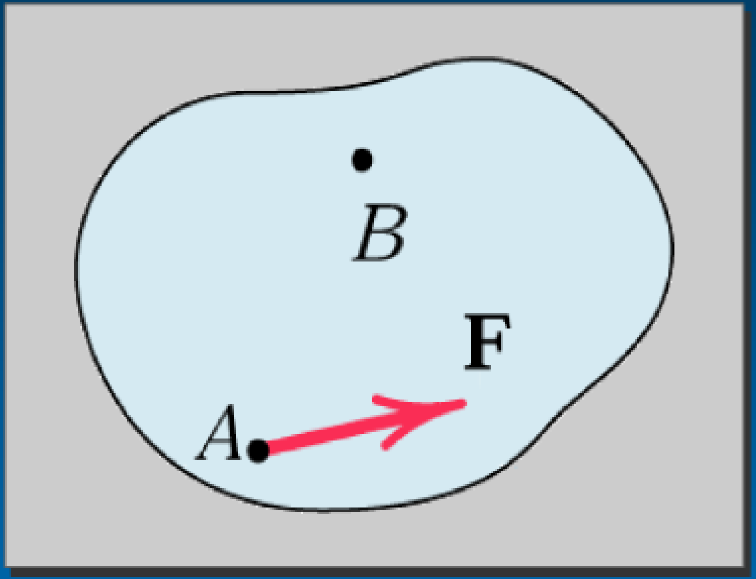
Moving force from point *A* to point *B*



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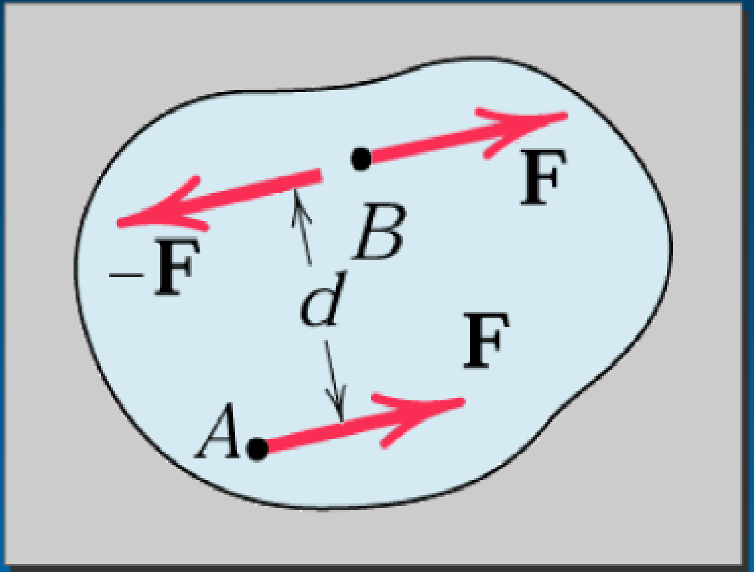
Moving force from point *A* to point *B*

F and –**F** are added at point *B*



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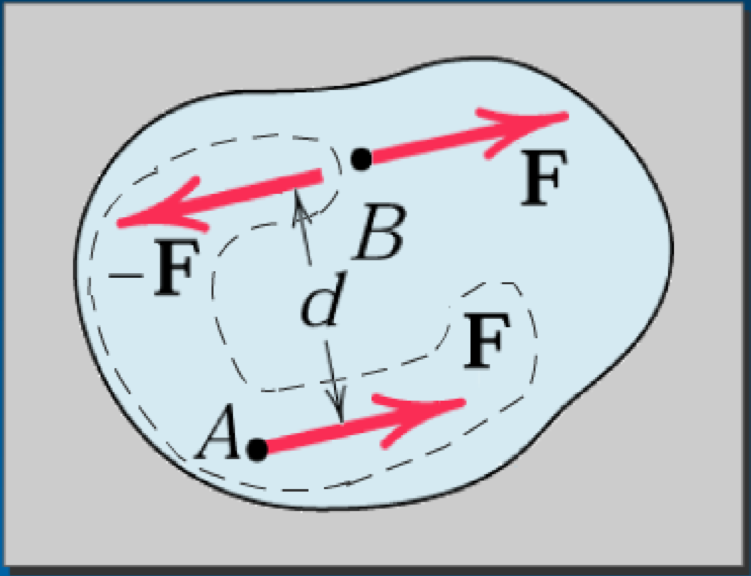
Moving force from point *A* to point *B*

F and –**F** form a couple.



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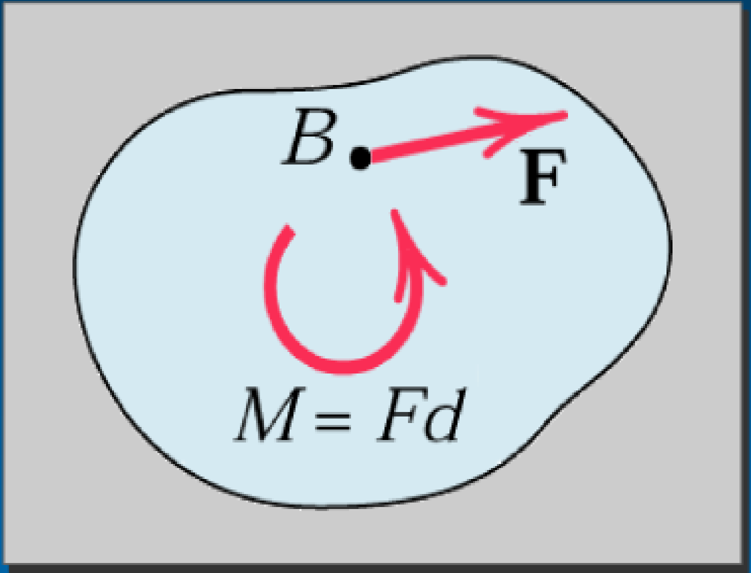
Moving force from point A to point B

$M = Fd$ is the moment associated with the couple **F** and –**F** .



The force has been moved to B, with the "penalty" of the added couple **M**.

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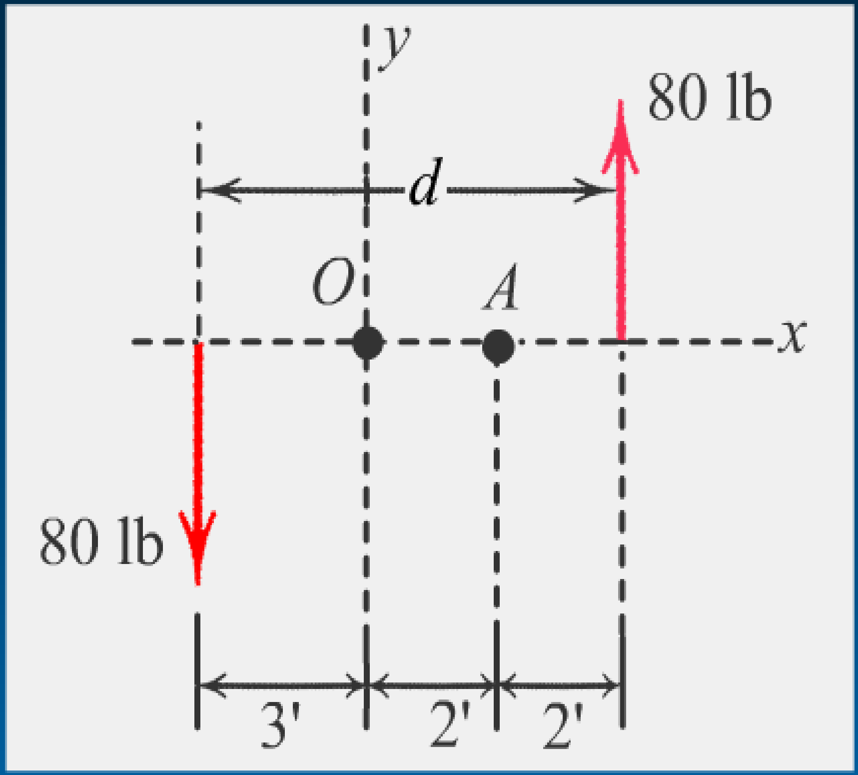

Determine the moment associated with the two 80-lb forces

(a) about point *O*

(b) about point *A*

Solution

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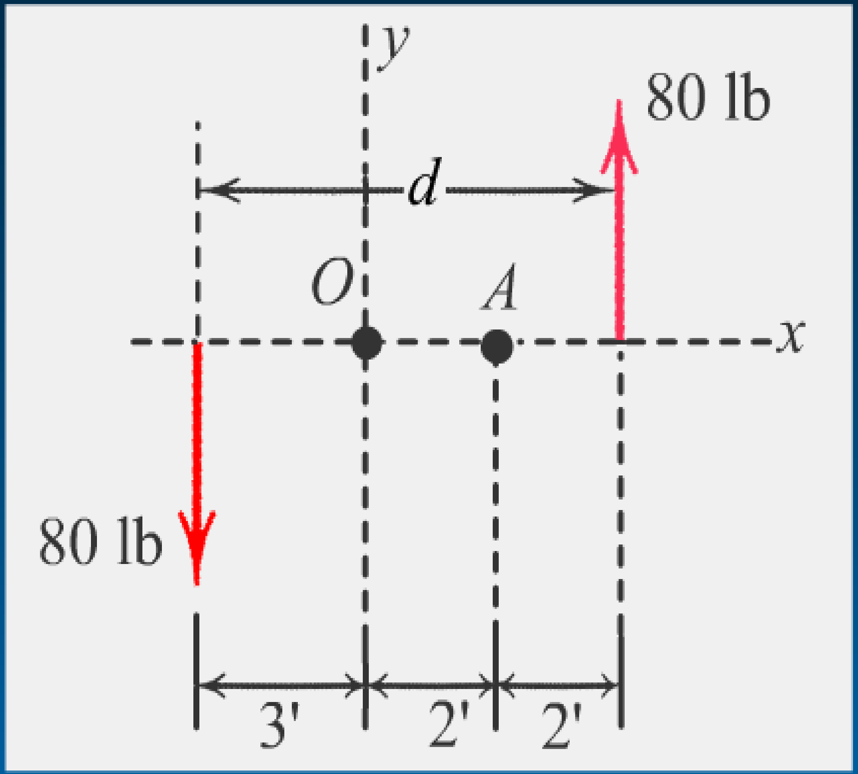

Determine the moment associated with the two 80-lb forces

(a) about point O

(b) about point A

Solution

(a) $M_o = 80(3) + 80(4) = 560$ lb-ft CCW

(b) $M_o = 80(5) + 80(2) = 560$ lb-ft CCW

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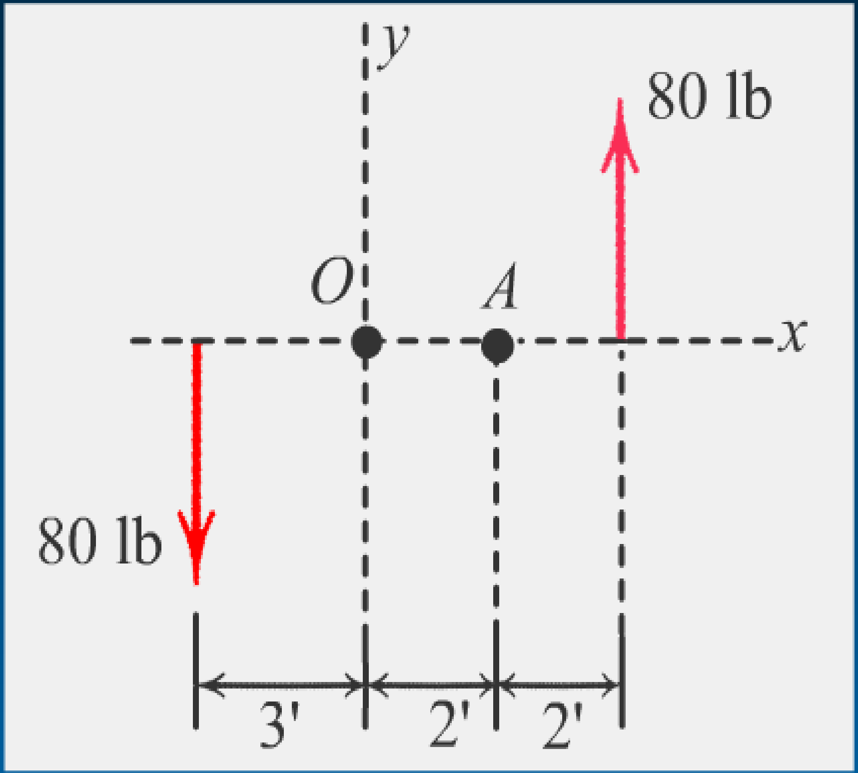

Note that the two results are identical. Recall that the moment associated with a couple is independent of the reference point (moment center). The value of the moment is just

$$M_o = Fd = 80(7) = 560 \text{ lb-ft}$$

where *d* is the perpendicular distance between the lines of action of the two forces.

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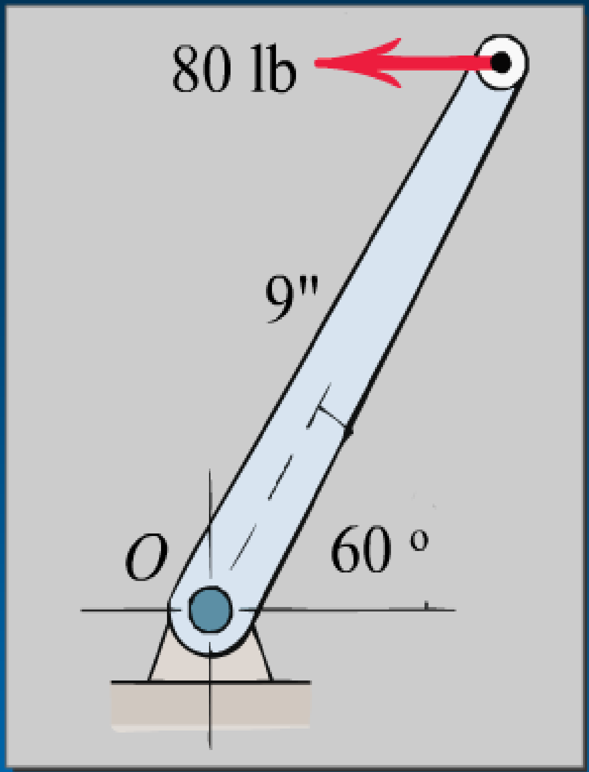

Example

Replace the horizontal 80-lb force acting on the lever by an equivalent system consisting of a force at *O* and a couple.

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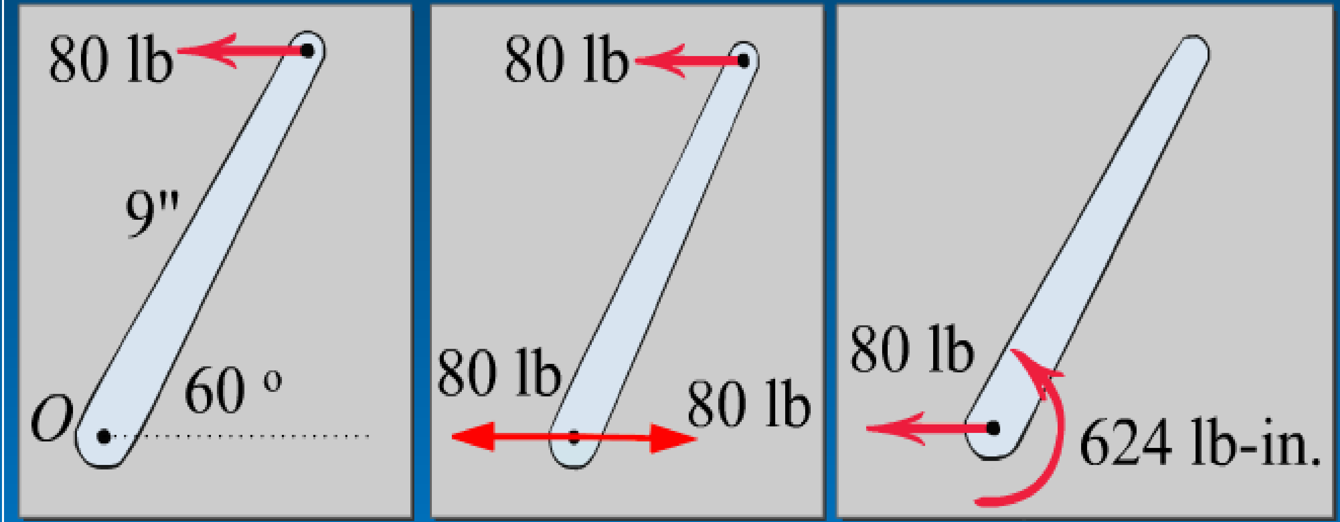
Solution:

We apply two-equal and opposite 80-lb forces at *O*
And identify the counterclockwise couple

$M = Fd = 80\ (\ 9 \sin 60^{o}\ ) = 624$ lb-in.



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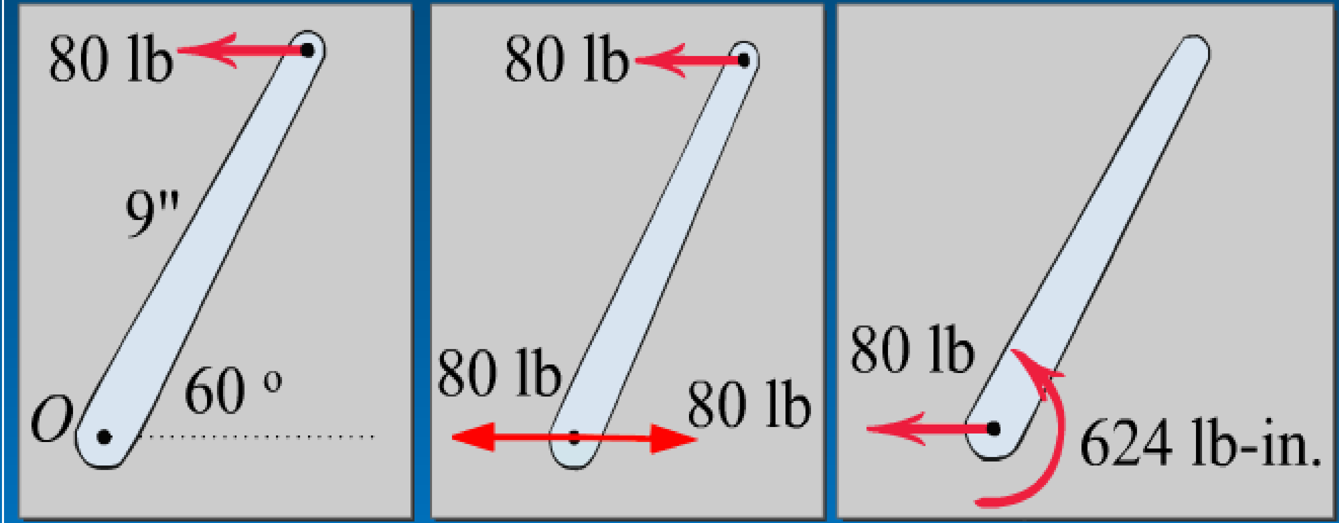
Solution:

We apply two-equal and opposite 80-lb forces at *O*

And identify the counterclockwise couple

$M = Fd = 80\ (\ 9 \sin 60^{o}\ ) = 624$ lb-in.

Thus, the original force is equivalent to the 80-lb force at *O* and the 624-lb-in. couple as shown in the final figure.



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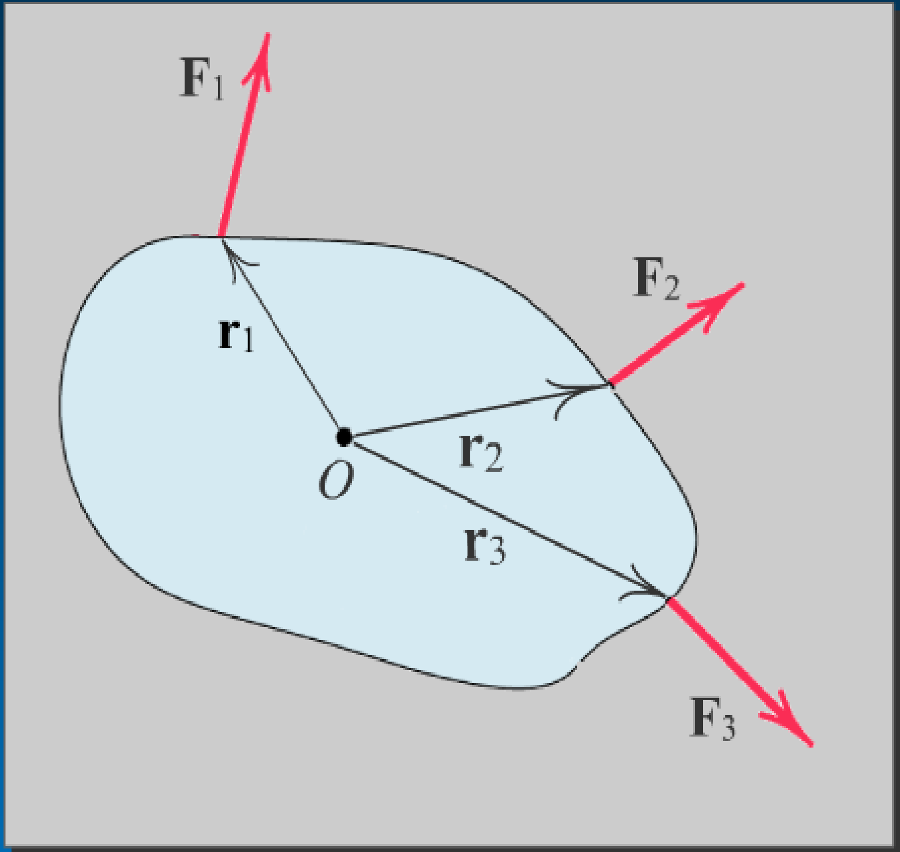
Resultants

We now reduce a system of forces and couples to the simplest possible representation, which is called the *resultant* of the original system.



Next

Click "Next" to move forces

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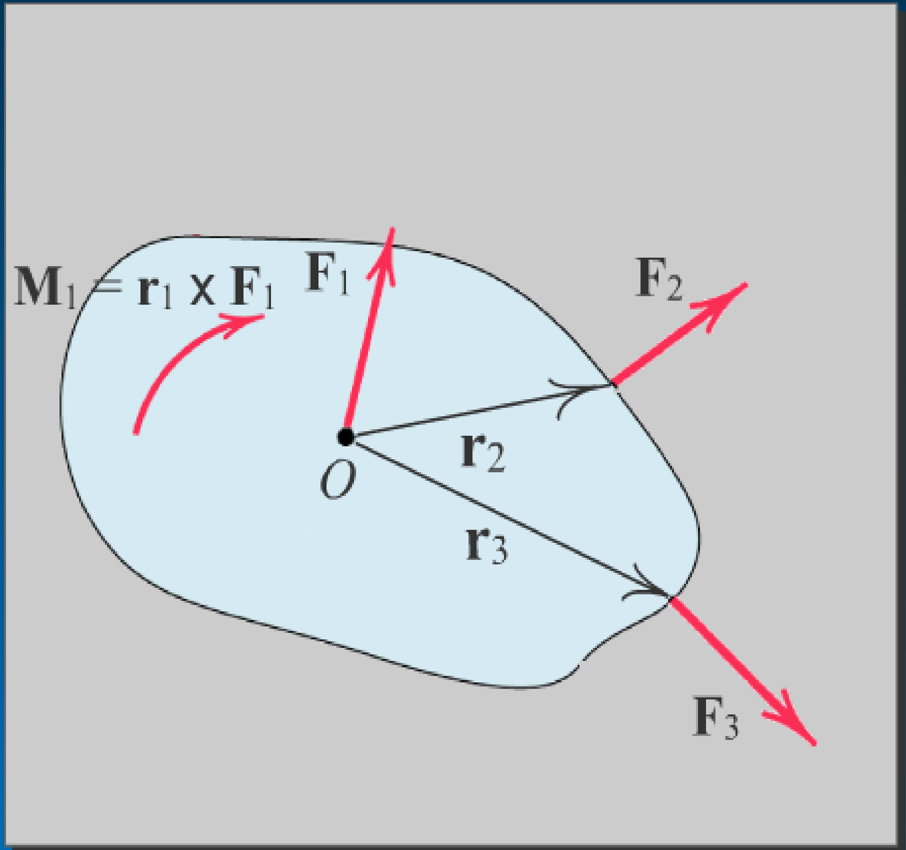
$\mathbf{F}_1$ moved to O.



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Click "Next" to move forces

1/1

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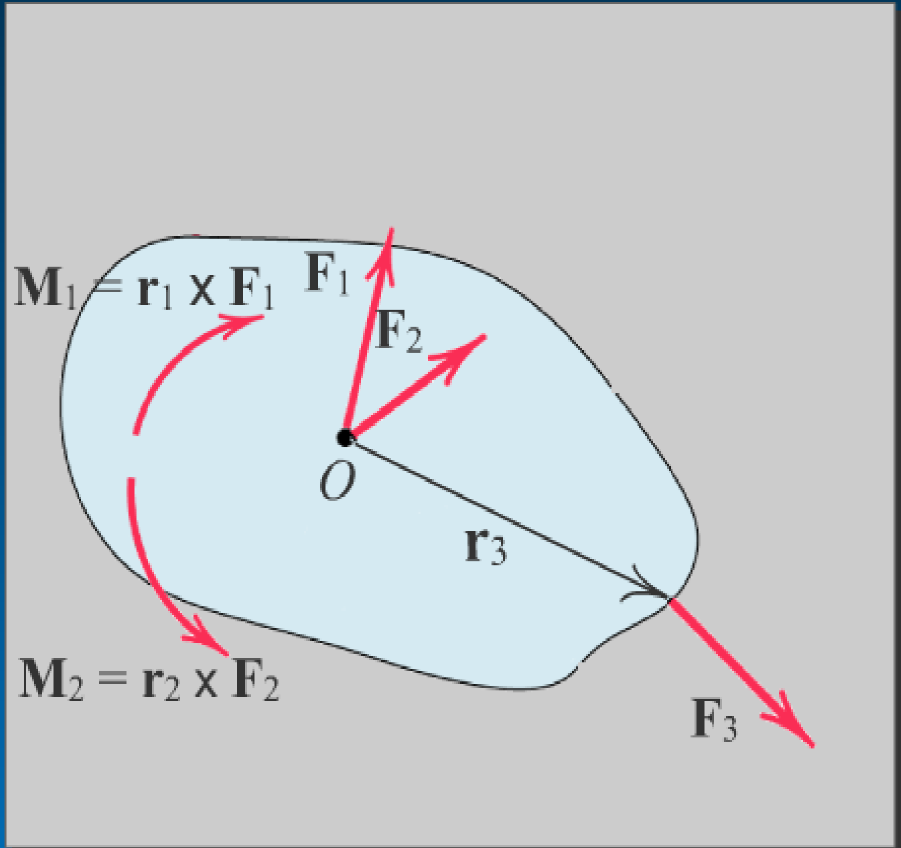
F_1, F_2 moved to *O*.



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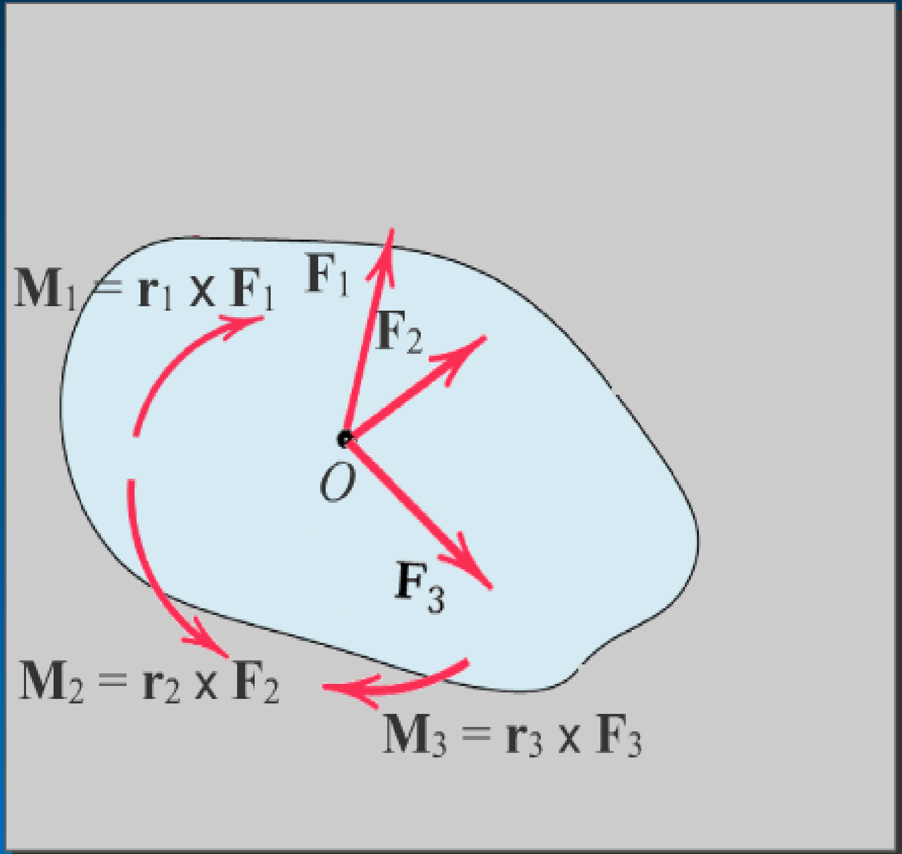
$\mathbf{F}_1$, $\mathbf{F}_2$, $\mathbf{F}_3$ moved to O.



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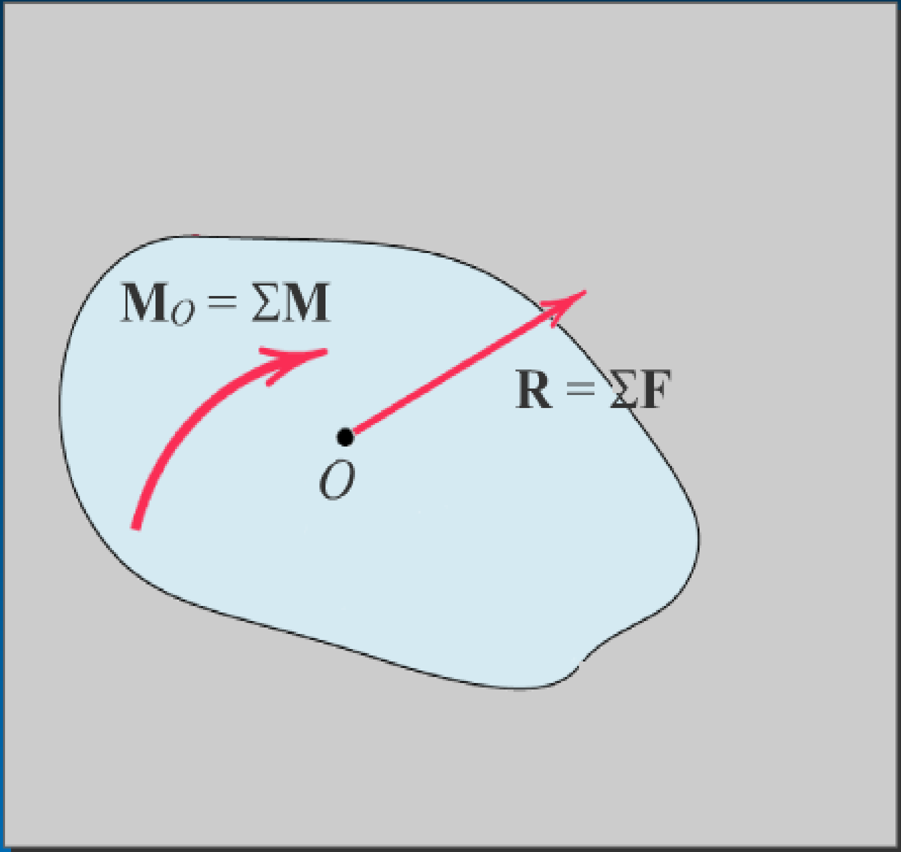
$\mathbf{F}_1$, $\mathbf{F}_2$, and $\mathbf{F}_3$ added to obtain $\mathbf{R}$.
$\mathbf{M}_1$, $\mathbf{M}_2$, and $\mathbf{M}_3$ added to obtain $\mathbf{M}_O$.



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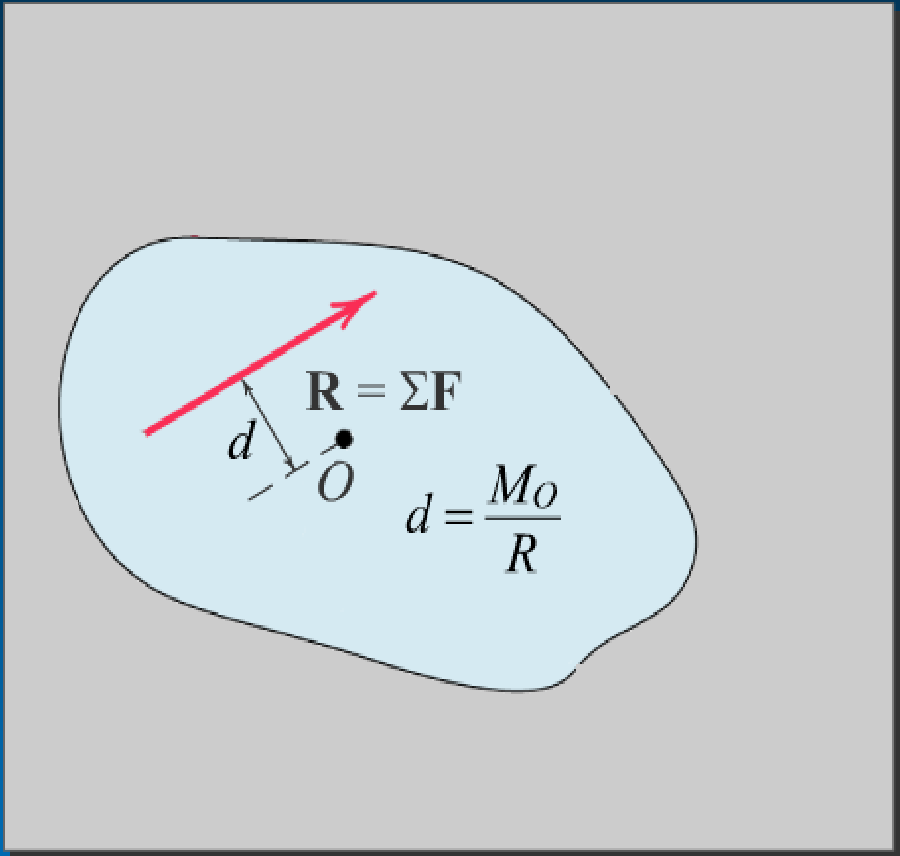
Force-couple system reduced to a single force.



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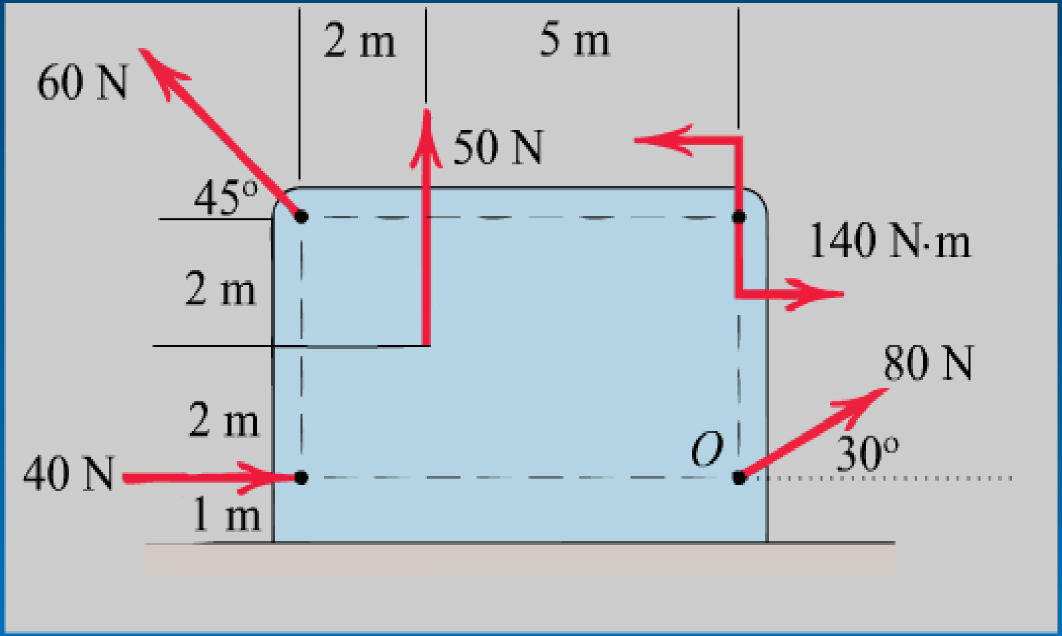
Example

Determine the resultant of the four forces and one couple which act on the plate shown.



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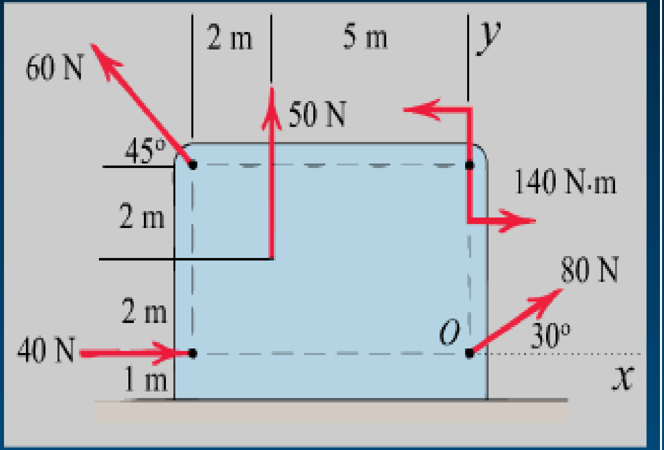
Solution

Point O is selected as a convenient point for the force couple system that is to represent the given system.



$$[R_x = \Sigma F_x]$$

$$[R_y = \Sigma F_y]$$

$$\left[R = \sqrt{R_x^2 + R_y^2}\right]$$

$$\left[\theta = \tan^{-1}\frac{R_y}{R_x}\right]$$

$$[\circlearrowright + \; M_o = \Sigma(Fd)]$$

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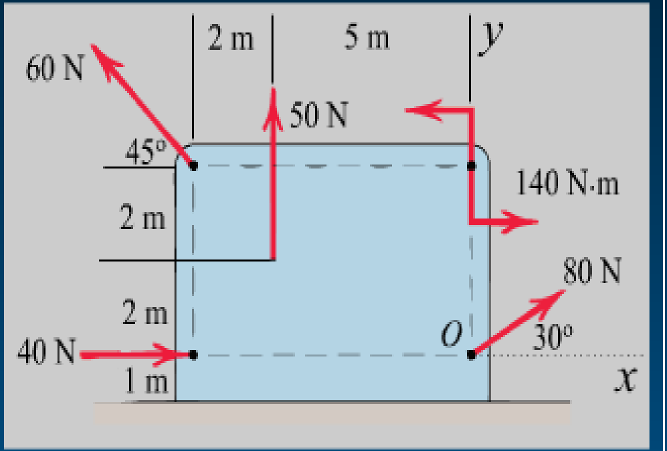

Solution

Point O is selected as a convenient point for the force couple system that is to represent the given system.

$[R_x = \Sigma F_x]$ $R_x = 40 + 80 \cos 30^o - 60 \cos 45^o = 66.9$ N

$[R_y = \Sigma F_y]$

$[R = \sqrt{R_x^2 + R_y^2}]$

$$\left[\theta = \tan^{-1} \frac{R_y}{R_x}\right]$$

$[+) M_o = \Sigma(Fd)]$

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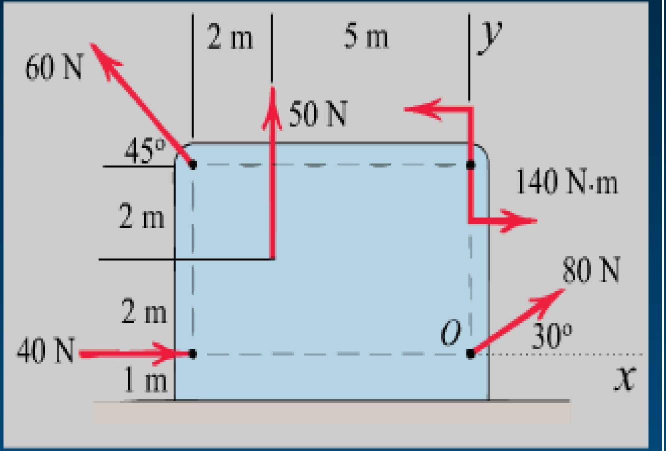
Solution

Point *O* is selected as a convenient point for the force couple system that is to represent the given system.



$[R_x = \Sigma F_x]$ $R_x = 40 + 80 \cos 30° - 60 \cos 45° = 66.9$ N

$[R_y = \Sigma F_y]$ $R_y = 50 + 80 \sin 30° + 60 \sin 45° = 132.4$ N

$$\left[R = \sqrt{R_x^2 + R_y^2}\right]$$

$$\left[\theta = \tan^{-1} \frac{R_y}{R_x}\right]$$

$$[\circlearrowleft\!+\ M_o = \Sigma(Fd)]$$

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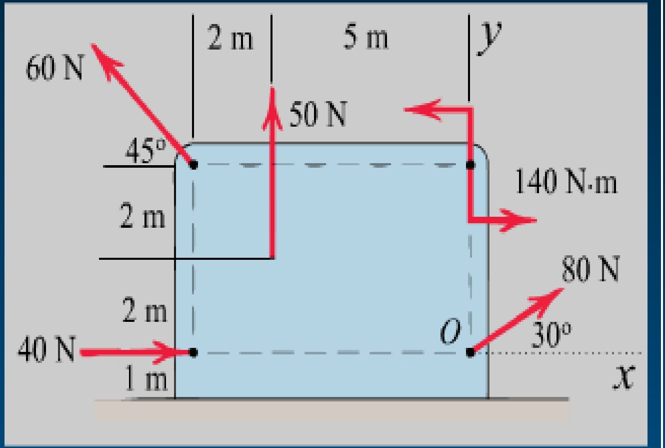
Solution

Point O is selected as a convenient point for the force couple system that is to represent the given system.



$$[R_x = \Sigma F_x] \quad R_x = 40 + 80\cos 30^\circ - 60\cos 45^\circ = 66.9\text{ N}$$

$$[R_y = \Sigma F_y] \quad R_y = 50 + 80\sin 30^\circ + 60\sin 45^\circ = 132.4\text{ N}$$

$$\left[R = \sqrt{R_x^2 + R_y^2}\right] \quad R = \sqrt{(66.9)^2 + (132.4)^2} = 148.3\text{ N}$$

$$\left[\theta = \tan^{-1}\frac{R_y}{R_x}\right]$$

$$[\overset{+}{\circlearrowleft}\ M_o = \Sigma(Fd)]$$

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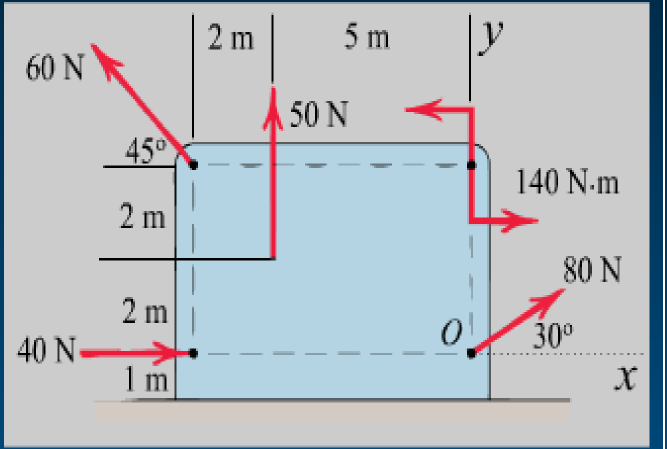
Solution

Point O is selected as a convenient point for the force couple system that is to represent the given system.



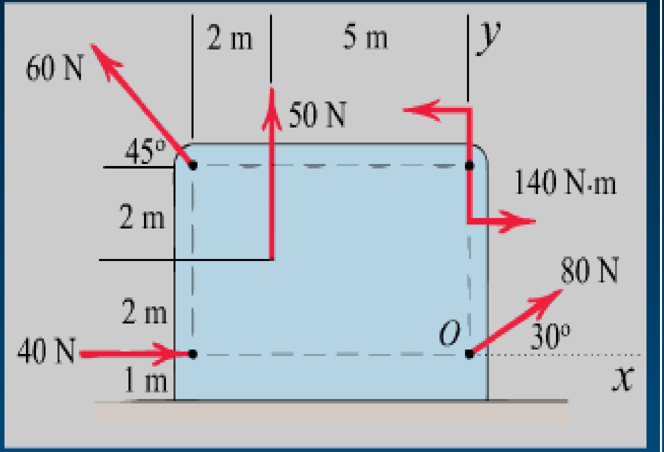
$$[R_x = \Sigma F_x] \qquad R_x = 40 + 80\cos 30^\circ - 60\cos 45^\circ = 66.9 \text{ N}$$

$$[R_y = \Sigma F_y] \qquad R_y = 50 + 80\sin 30^\circ + 60\sin 45^\circ = 132.4 \text{ N}$$

$$\left[R = \sqrt{R_x^2 + R_y^2}\right] \qquad R = \sqrt{(66.9)^2 + (132.4)^2} = 148.3 \text{ N}$$

$$\left[\theta = \tan^{-1}\frac{R_y}{R_x}\right] \qquad \theta = \tan^{-1}\frac{132.4}{66.9} = 63.2^\circ$$

$$[+\circlearrowleft\ M_o = \Sigma(Fd)]$$



Solution

Point O is selected as a convenient point for the force couple system that is to represent the given system.

$$[R_x = \Sigma F_x] \qquad R_x = 40 + 80 \cos 30^\circ - 60 \cos 45^\circ = 66.9 \text{ N}$$

$$[R_y = \Sigma F_y] \qquad R_y = 50 + 80 \sin 30^\circ + 60 \sin 45^\circ = 132.4 \text{ N}$$

$$\left[R = \sqrt{R_x^2 + R_y^2}\right] \qquad R = \sqrt{(66.9)^2 + (132.4)^2} = 148.3 \text{ N}$$

$$\left[\theta = \tan^{-1} \frac{R_y}{R_x}\right] \qquad \theta = \tan^{-1} \frac{132.4}{66.9} = 63.2^\circ$$

$$[(+\circlearrowright)\; M_o = \Sigma(Fd)] \quad M_o = 140 - 50(5) + 60 \cos 45^\circ(4) - 60 \sin 45^\circ(7)$$
$$= -237 \text{ N·m}$$

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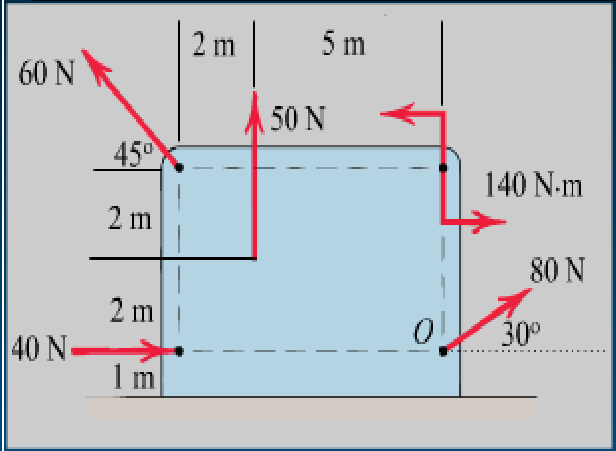
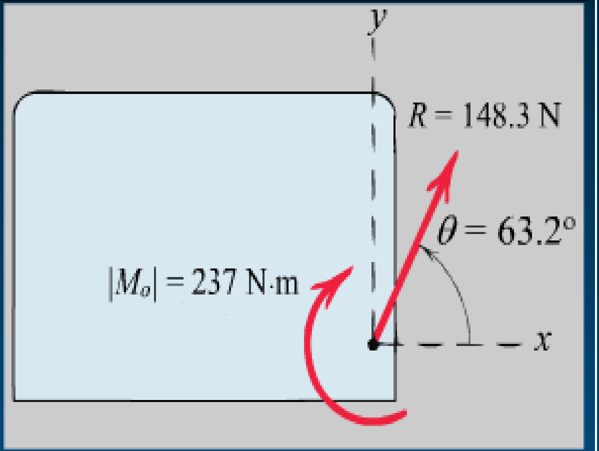


The force-couple system (**R** and **M**$_o$) is shown in the right figure.



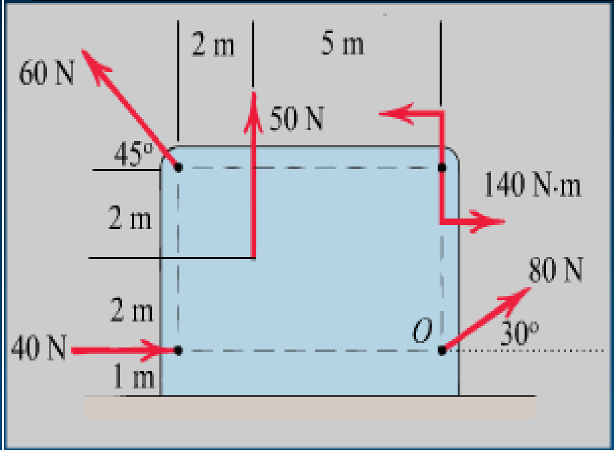
$[R_x = \Sigma F_x]$ $R_x = 40 + 80 \cos 30^\circ - 60 \cos 45^\circ = 66.9$ N

$[R_y = \Sigma F_y]$ $R_y = 50 + 80 \sin 30^\circ + 60 \sin 45^\circ = 132.4$ N

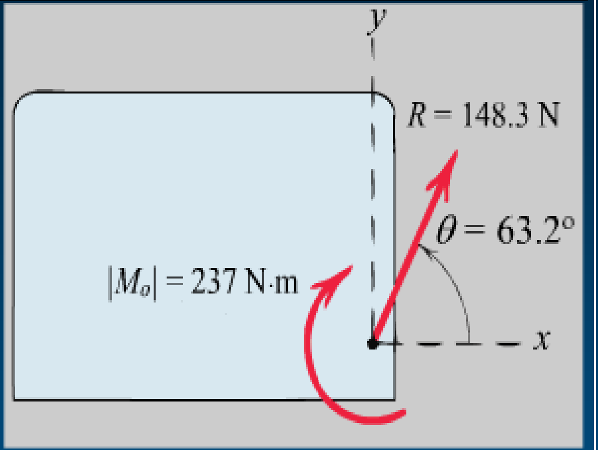
$[R = \sqrt{R_x^2 + R_y^2}]$ $R = \sqrt{(66.9)^2 + (132.4)^2} = 148.3$ N

$$\left[\theta = \tan^{-1}\frac{R_y}{R_x}\right] \qquad \theta = \tan^{-1}\frac{132.4}{66.9} = 63.2^\circ$$

$[+\circlearrowleft\, M_o = \Sigma(Fd)]$ $M_o = 140 - 50(5) + 60 \cos 45^\circ(4) - 60 \sin 45^\circ(7)$
$= -237$ N·m



The force-couple system (**R** and **M**$_o$) is shown in the right figure.



We now determine the final line of action of **R** such that **R** alone represents the original system.

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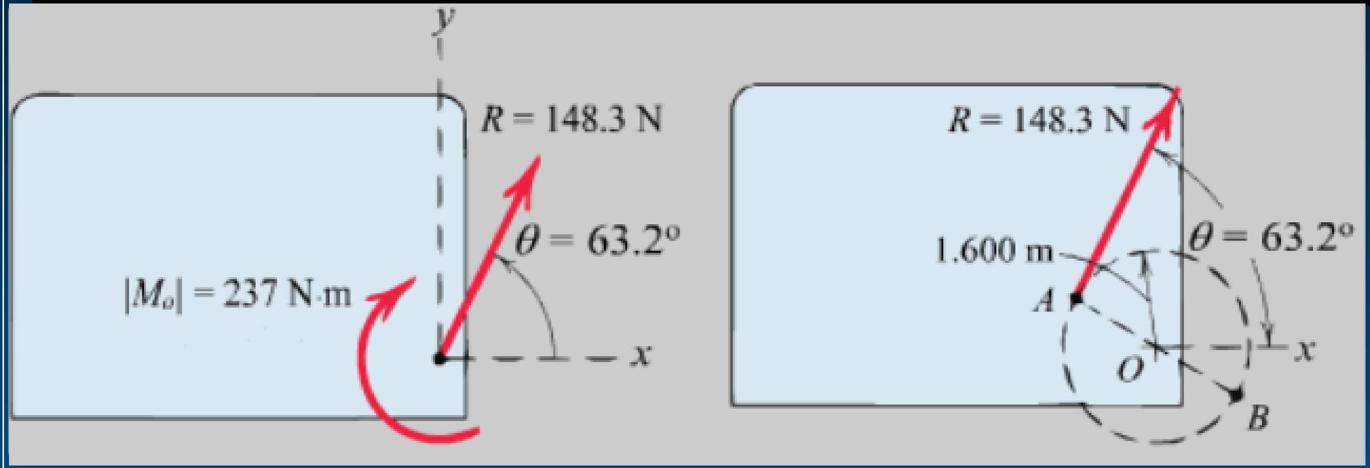

We now determine the final line of action of **R** such that **R** alone represents the original system.

$[Rd = |M_o|]$ $\quad 1.483d = 237 \quad d = 1.600$ m

Hence the resultant **R** may be applied at any point on the line which makes a 63.2° angle with the *x*-axis and is tangent at the point *A* to a circle of 1.6-m radius with center *O*, as shown in the right figure.

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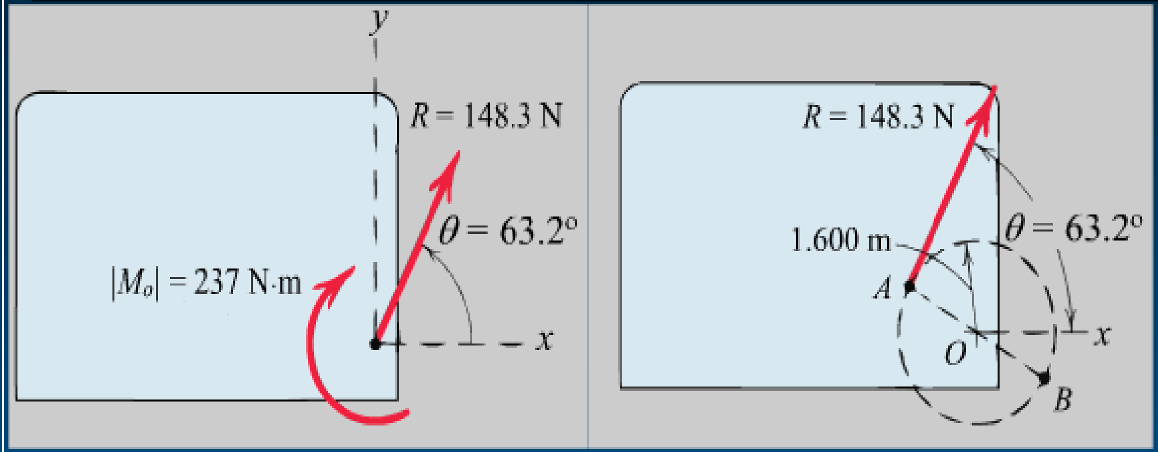

$[Rd = |M_o|]$ $\quad 1.483d = 237 \quad d = 1.600$ m

We apply the equation *Rd* = *M* in the absolute-value sence (ignoring the sign of M_o) and let the physics of the situation, as depicted in the left figure, dictate the final placement of **R**. Had M_o been counterclockwise, the correct line of action of **R** would have been the tangent at point *B*.

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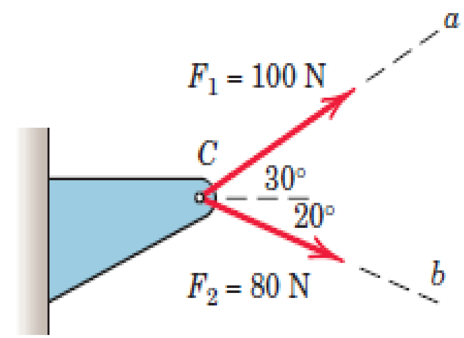
Sample Problem 2/4

Forces $\mathbf{F}_1$ and $\mathbf{F}_2$ act on the bracket as shown. Determine the projection F_b of their resultant $\mathbf{R}$ onto the b-axis.



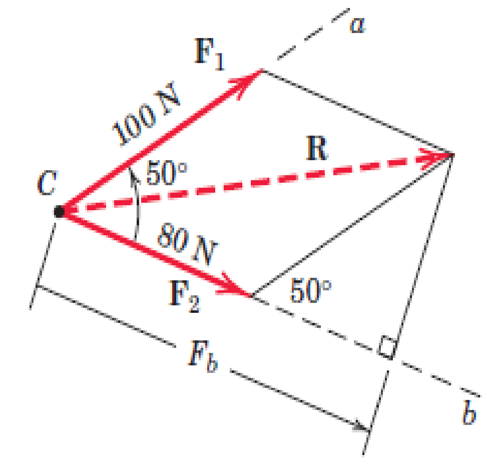
Solution. The parallelogram addition of $\mathbf{F}_1$ and $\mathbf{F}_2$ is shown in the figure. Using the law of cosines gives us

$$R^2 = (80)^2 + (100)^2 - 2(80)(100)\cos 130^\circ \qquad R = 163.4 \text{ N}$$

The figure also shows the orthogonal projection $\mathbf{F}_b$ of $\mathbf{R}$ onto the b-axis. Its length is

$$F_b = 80 + 100 \cos 50^\circ = 144.3 \text{ N} \qquad \textit{Ans.}$$

Note that the components of a vector are in general not equal to the projections of the vector onto the same axes. If the a-axis had been perpendicular to the b-axis, then the projections and components of $\mathbf{R}$ would have been equal.



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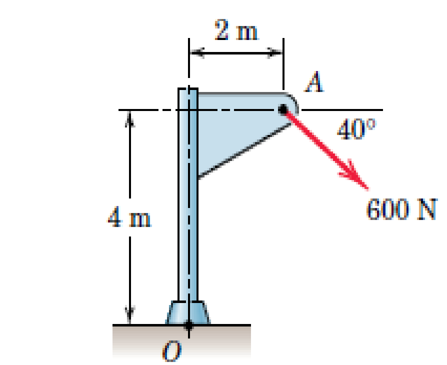
Sample Problem 2/5

Calculate the magnitude of the moment about the base point O of the 600-N force in five different ways.



Solution. *(I)* The moment arm to the 600-N force is

$$d = 4 \cos 40° + 2 \sin 40° = 4.35 \text{ m}$$

① By $M = Fd$ the moment is clockwise and has the magnitude

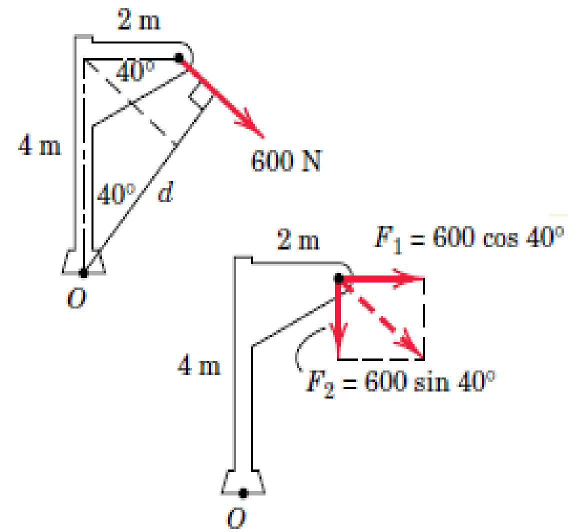
$$M_O = 600(4.35) = 2610 \text{ N·m} \qquad \textit{Ans.}$$

(II) Replace the force by its rectangular components at A

$$F_1 = 600 \cos 40° = 460 \text{ N}, \qquad F_2 = 600 \sin 40° = 386 \text{ N}$$

By Varignon's theorem, the moment becomes

② $$M_O = 460(4) + 386(2) = 2610 \text{ N·m} \qquad \textit{Ans.}$$



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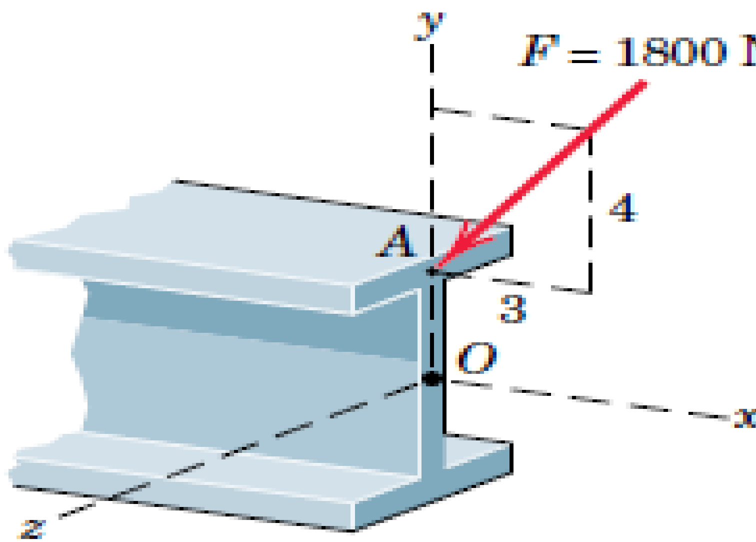
2/6 The 1800-N force **F** is applied to the end of the I-beam. Express **F** as a vector using the unit vectors **i** and **j**.



2/6	$\mathbf{F} = 1800\left(-\frac{3}{5}\mathbf{i} - \frac{4}{5}\mathbf{j}\right) = -1080\mathbf{i} - 1440\mathbf{j}$ N

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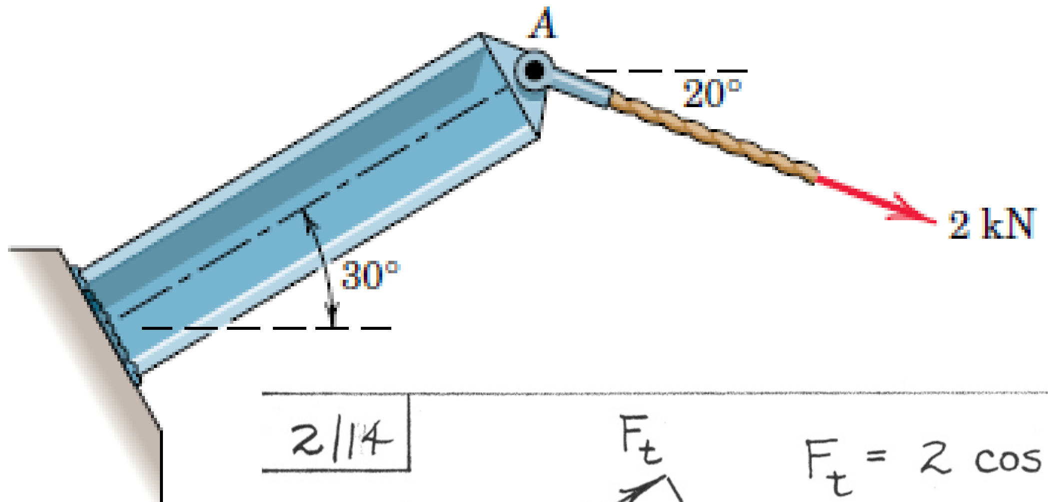
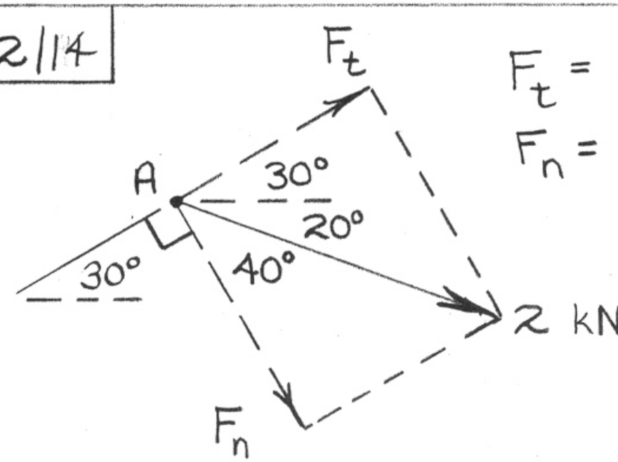
2/14 To satisfy design limitations it is necessary to determine the effect of the 2-kN tension in the cable on the shear, tension, and bending of the fixed I-beam. For this purpose replace this force by its equivalent of two forces at A, F_t parallel and F_n perpendicular to the beam. Determine F_t and F_n.



2/14



$F_t = 2 \cos 50° = \underline{1.286 \text{ kN}}$

$F_n = 2 \sin 50° = \underline{1.532 \text{ kN}}$

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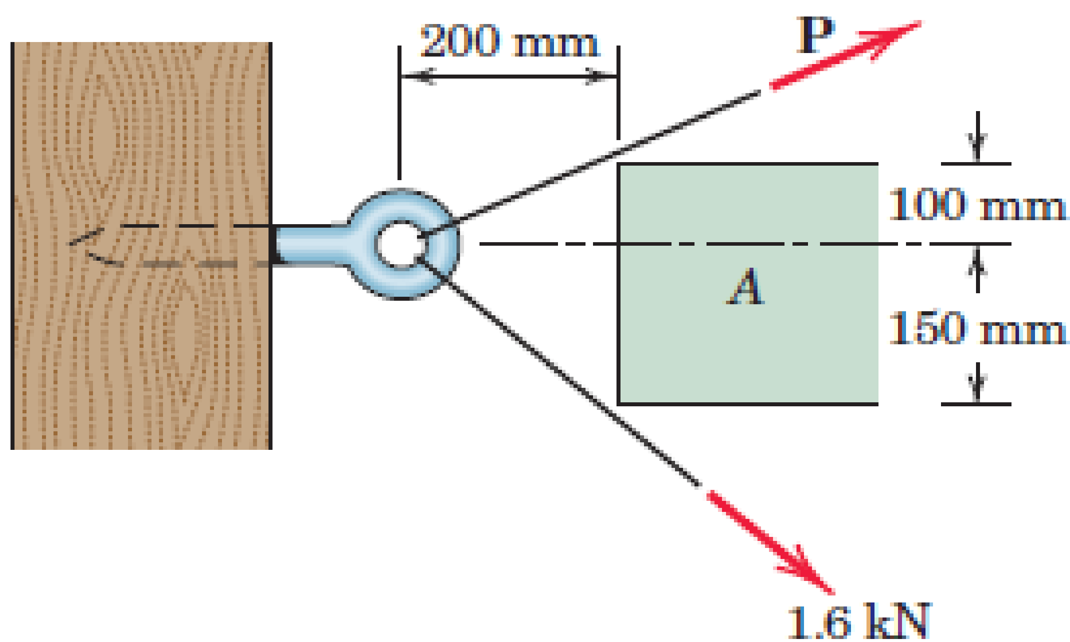
2/20 It is desired to remove the spike from the timber by applying force along its horizontal axis. An obstruction *A* prevents direct access, so that two forces, one 1.6 kN and the other **P**, are applied by cables as shown. Compute the magnitude of **P** necessary to ensure a resultant **T** directed along the spike. Also find *T*.



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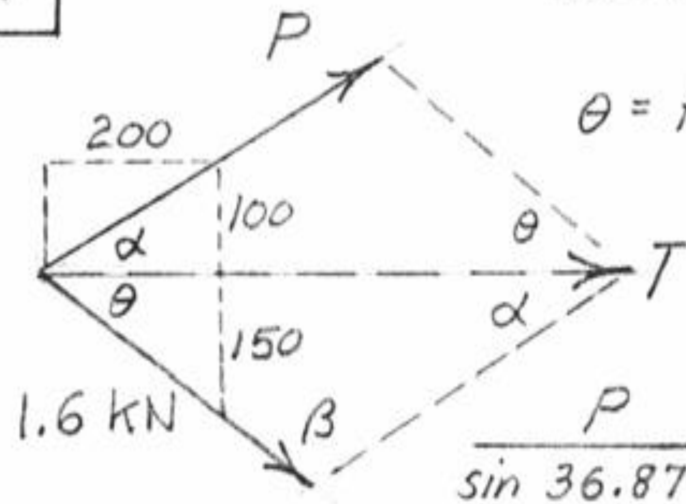
2/20



$$\alpha = \tan^{-1}\frac{100}{200} = 26.57^\circ$$

$$\theta = \tan^{-1}\frac{150}{200} = 36.87^\circ$$

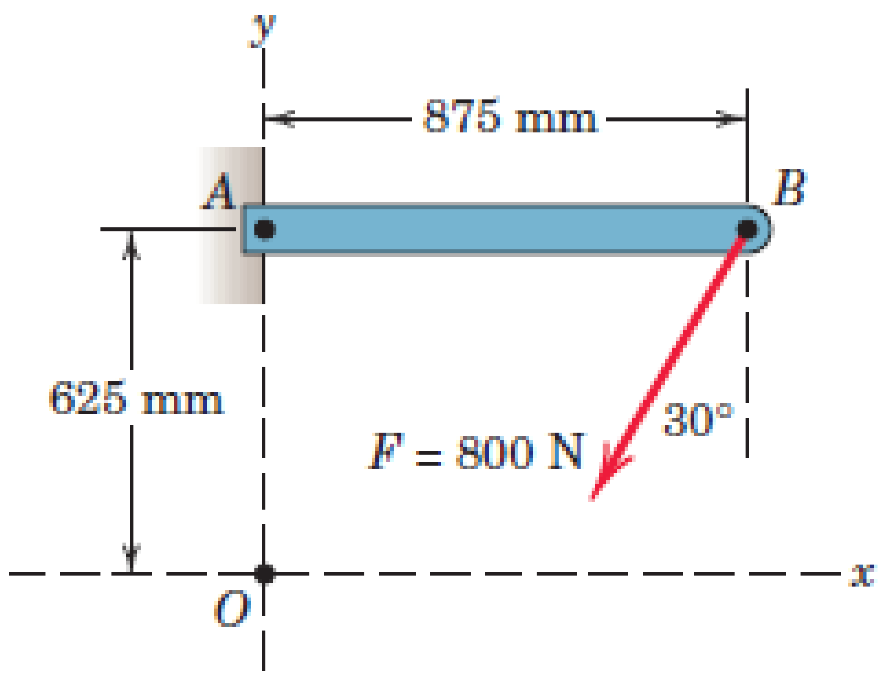
$$\beta = 180 - (\alpha + \theta) = 116.57^\circ$$

$$\frac{P}{\sin 36.87^\circ} = \frac{1.6}{\sin 26.57}$$

$$P = 1.6\,\frac{0.6}{0.4472} = \underline{2.15\ \text{kN}}$$

$$\frac{T}{\sin 116.57^\circ} = \frac{1.6}{\sin 26.57^\circ}, \quad T = 1.6\,\frac{0.8944}{0.4472} = \underline{3.20\ \text{kN}}$$

2/30 Determine the moment of the 800-N force about point A and about point O.



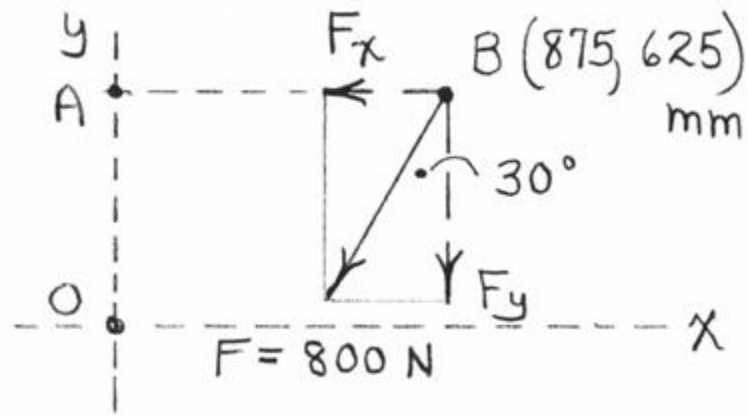
2/30

$|F_x| = 800 \sin 30°$

$= 400 \text{ N}$

$|F_y| = 800 \cos 30°$

$= 693 \text{ N}$



$\circlearrowright M_A = 693(0.875) = \underline{606 \text{ N}\cdot\text{m} \quad \text{CW}}$

$\circlearrowright M_O = 693(0.875) - 400(0.625)$

$= \underline{356 \text{ N}\cdot\text{m} \quad \text{CW}}$

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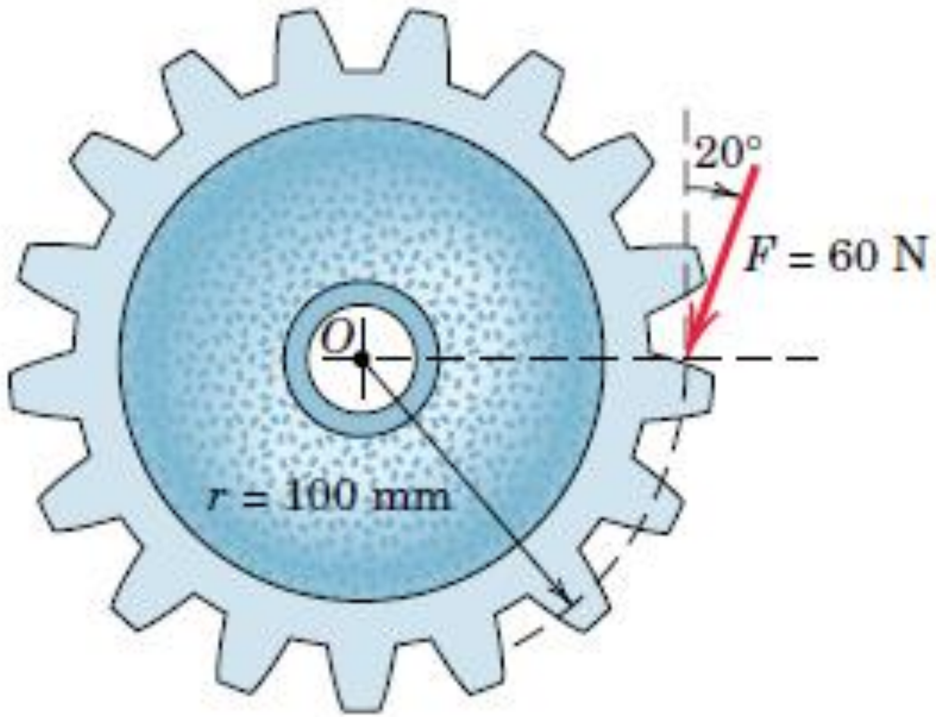
2/35 A force **F** of magnitude 60 N is applied to the gear. Determine the moment of **F** about point O.

Ans. $M_O = 5.64\ \text{N}\cdot\text{m CW}$



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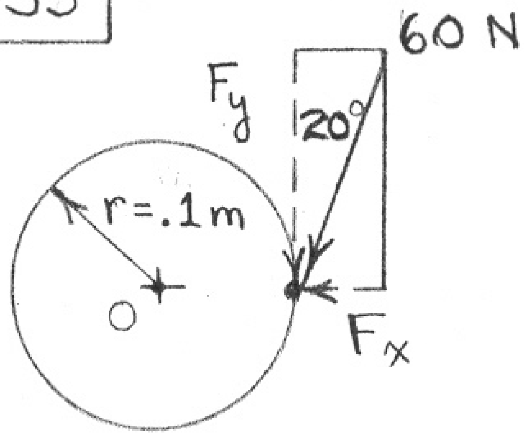
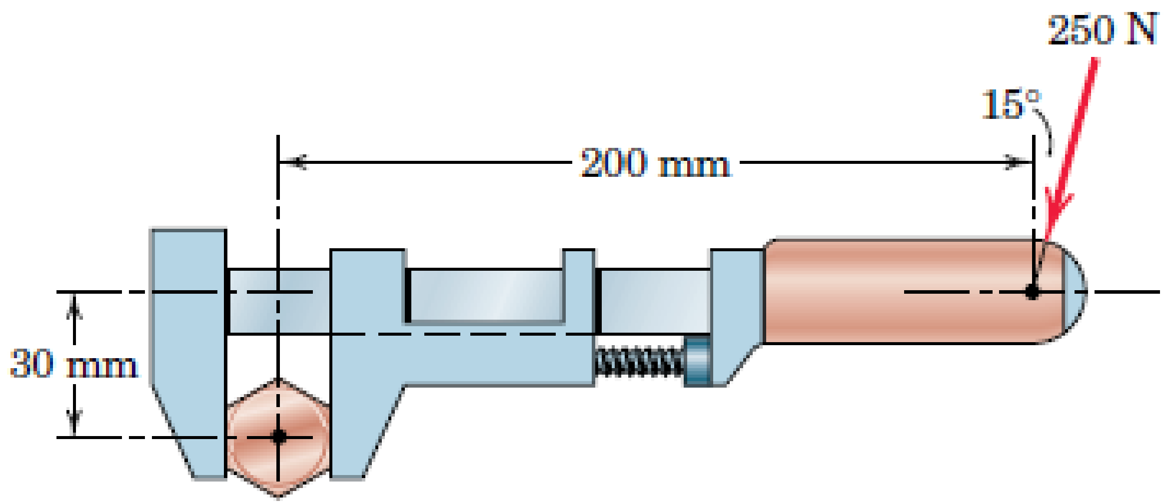


$$+\circlearrowright M_O = r F_y$$
$$= (0.1)(60 \cos 20°)$$
$$= \underline{5.64 \text{ N}\cdot\text{m}}$$

2/36 Calculate the moment of the 250-N force on the handle of the monkey wrench about the center of the bolt.



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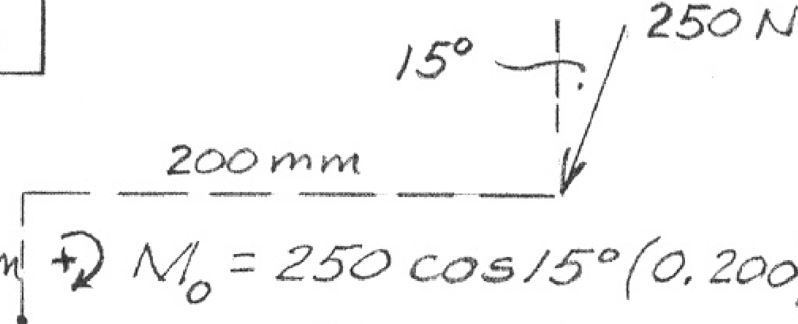
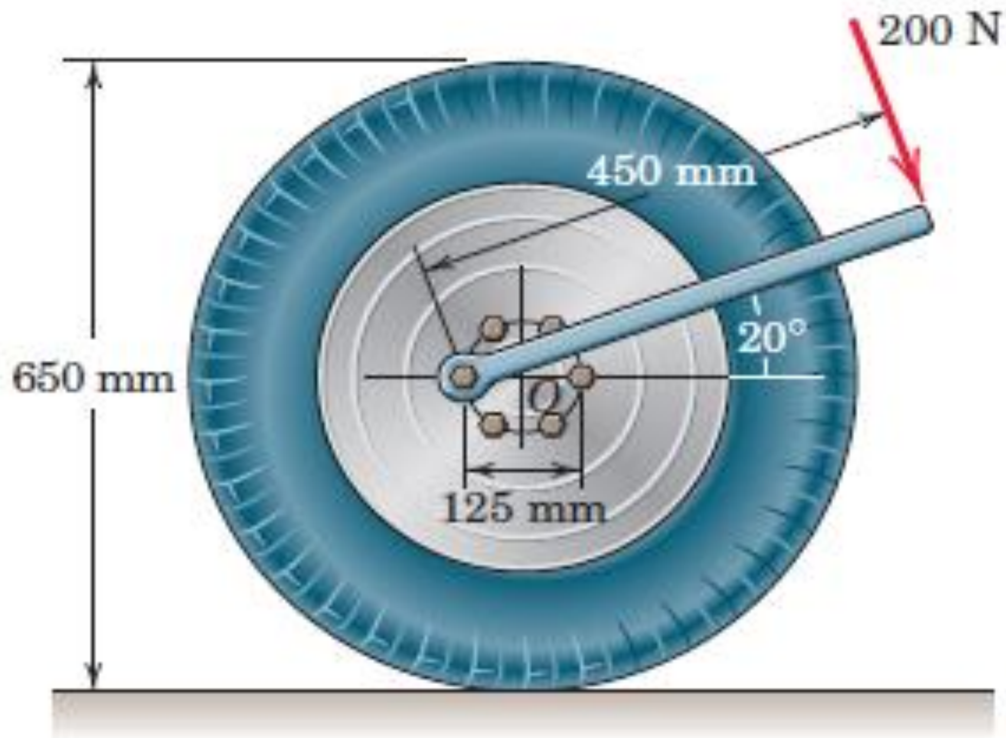


$$+\circlearrowright M_O = 250\cos 15^\circ(0.200) - 250\sin 15^\circ(0.030)$$

$$= 48.30 - 1.941 = \underline{46.4\ \text{N}\cdot\text{m}}$$

2/42 A force of 200 N is applied to the end of the wrench to tighten a flange bolt which holds the wheel to the axle. Determine the moment M produced by this force about the center O of the wheel for the position of the wrench shown.



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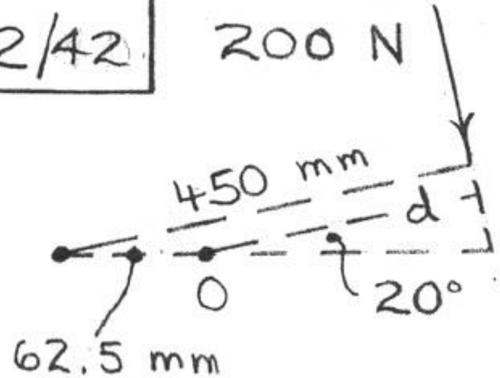
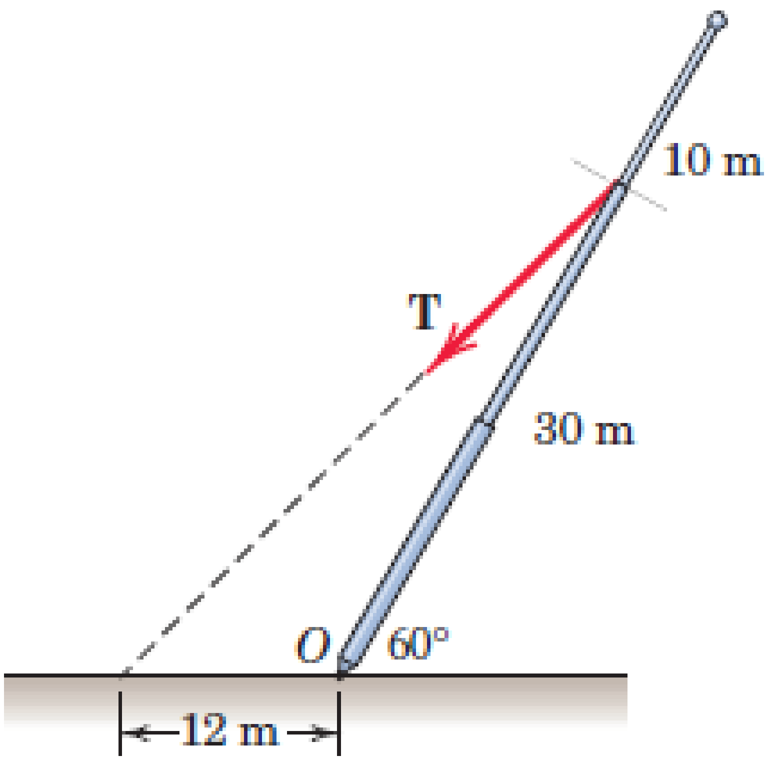


$$d = 450 - 62.5 \cos 20^\circ$$
$$= 391 \text{ mm}$$

$$\circlearrowleft + \; M = Fd = 200(0.391)$$
$$= \underline{78.3 \text{ N·m}}$$

2/45 In raising the pole from the position shown, the tension T in the cable must supply a moment about O of $72\ \text{kN}\cdot\text{m}$. Determine T.

Ans. $T = 8.65\ \text{kN}$



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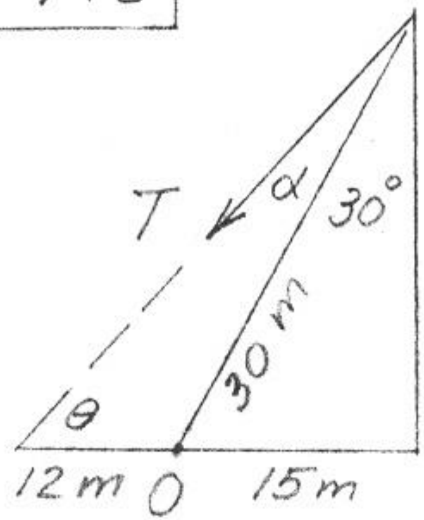

$$\theta = \tan^{-1}\frac{30(0.866)}{12+15} = 43.90^\circ$$

$$\alpha = 90^\circ - (30^\circ + 43.90^\circ) = 16.10^\circ$$

$$M_O = 72 \text{ kN}\cdot\text{m}$$

$$= T \sin 16.10^\circ (30) = 8.32T$$

$$T = \frac{72}{8.32} = \underline{8.65 \text{ kN}}$$

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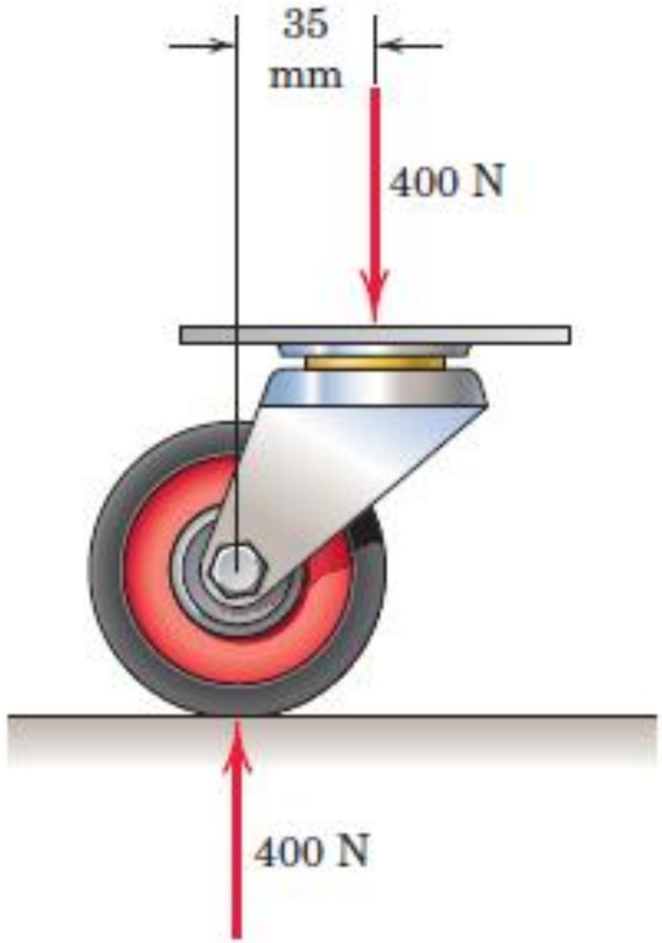
2/57 The caster unit is subjected to the pair of 400-N forces shown. Determine the moment associated with these forces.

Ans. $M = 14\ \text{N}\cdot\text{m CW}$



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$$\overset{+}{\circlearrowright} \quad M = Fd = 400(0.035)$$

$$= \underline{14 \text{ N}\cdot\text{m} \quad CW}$$

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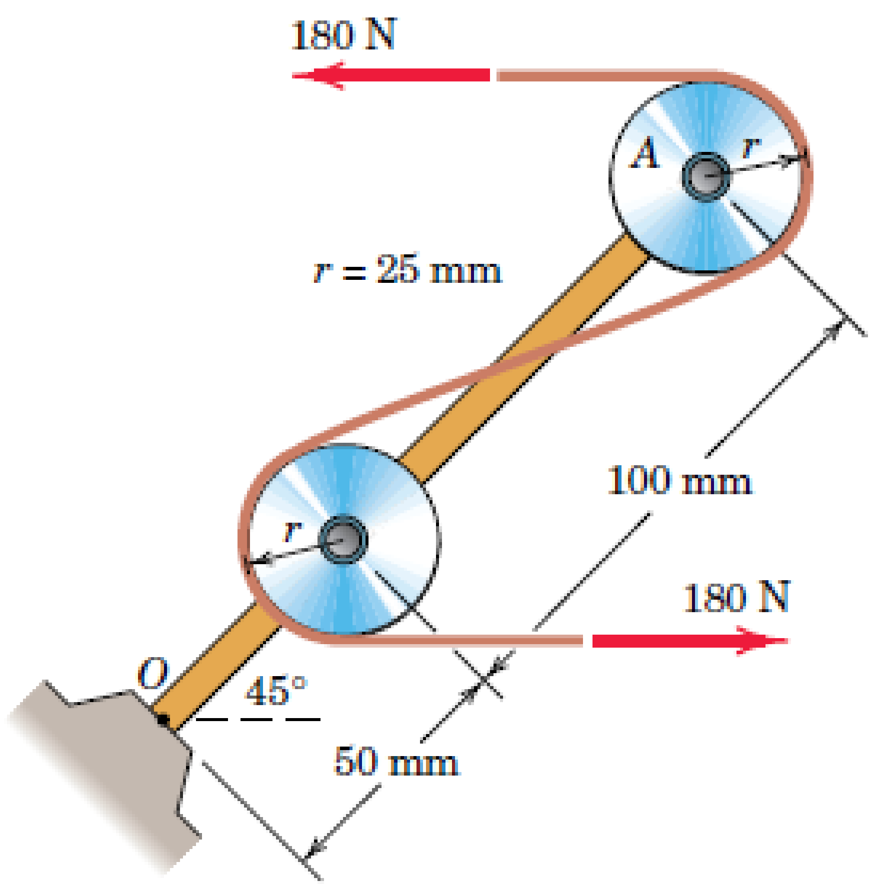
2/71 The system consisting of the bar OA, two identical pulleys, and a section of thin tape is subjected to the two 180-N tensile forces shown in the figure. Determine the equivalent force–couple system at point O.

Ans. $M = 21.7\ \text{N}\cdot\text{m CCW}$



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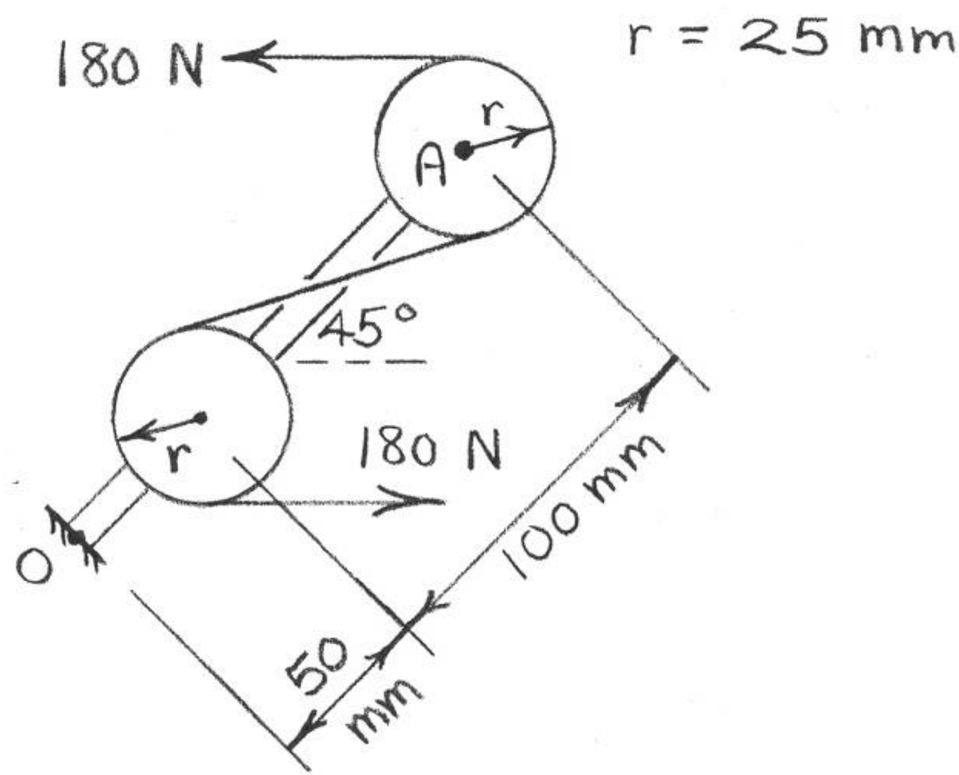

The system at O is a couple.

$$\circlearrowleft + M = Fd = 180(100 \sin 45° + 25 + 25)$$

$$= 21\,700 \text{ N·mm or } \underline{21.7 \text{ N·m CCW}}$$

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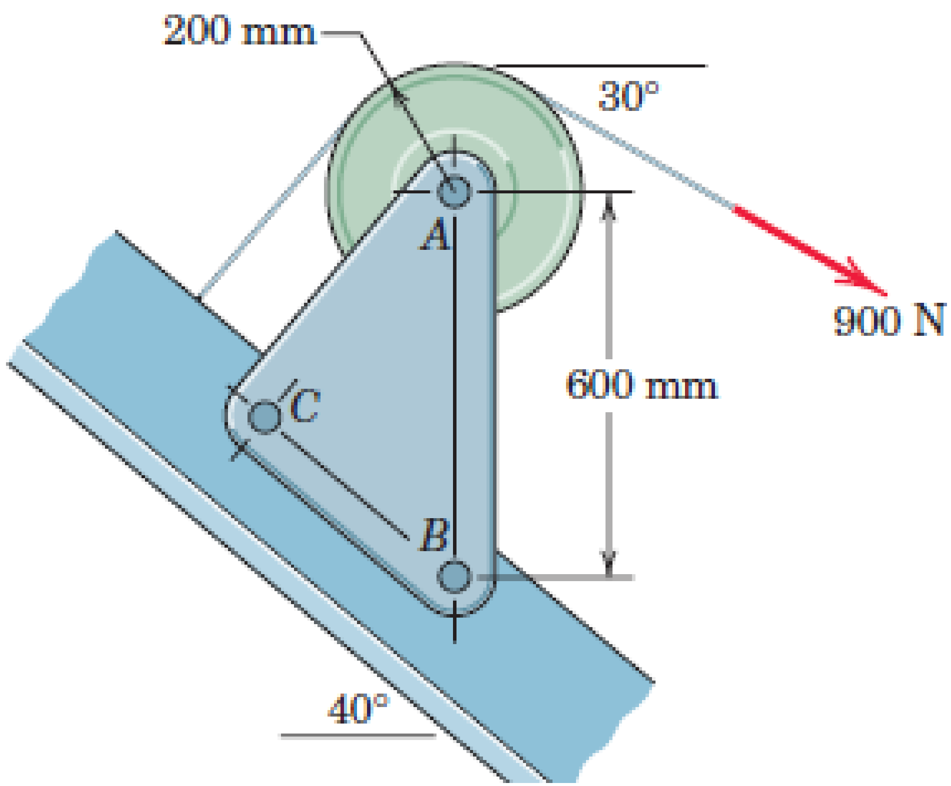
2/72 Calculate the moment M_B of the 900-N force about the bolt at B. Simplify your work by first replacing the force by its equivalent force–couple system at A.



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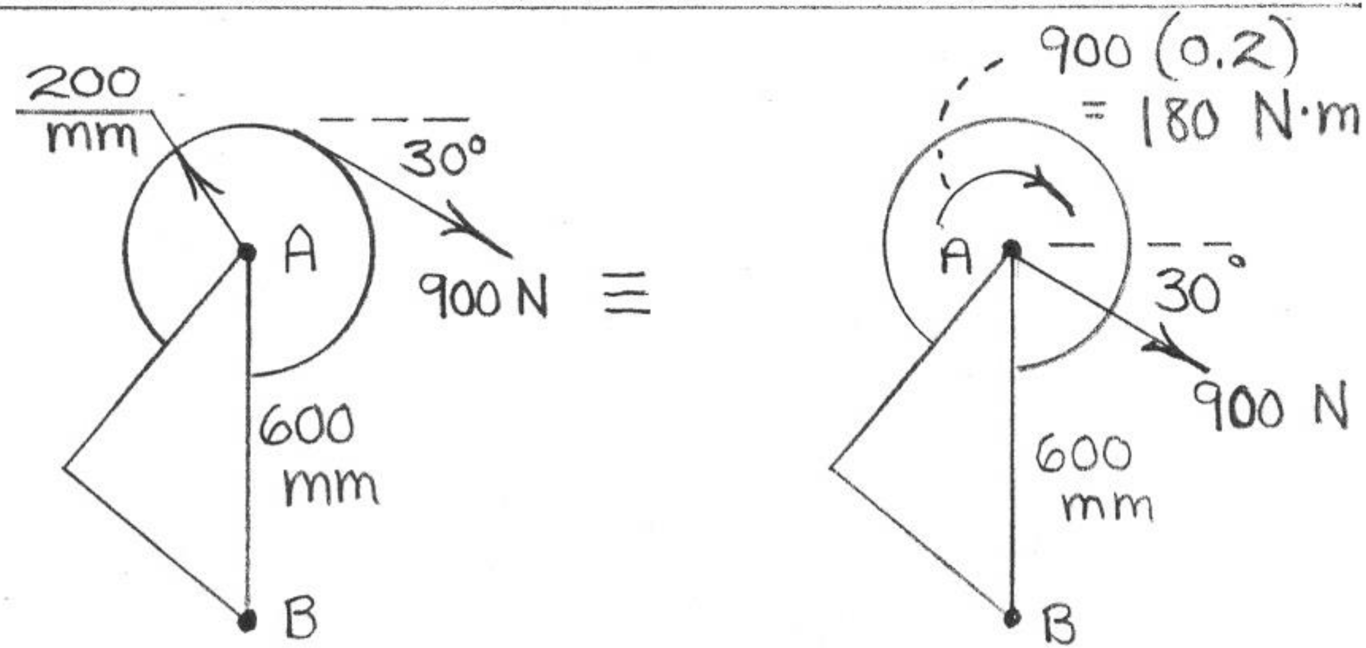
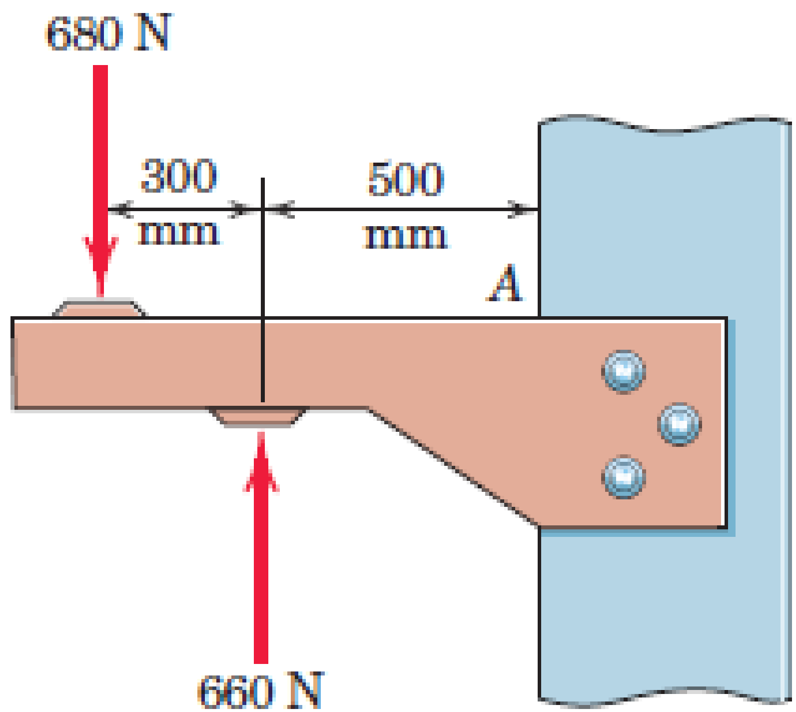


$$+\circlearrowright\ M_B = 180 + 900\cos 30^\circ (0.6) = \underline{648 \text{ N}\cdot\text{m CW}}$$

2/81 Where does the resultant of the two forces act?

Ans. 10.70 m to the left of *A*



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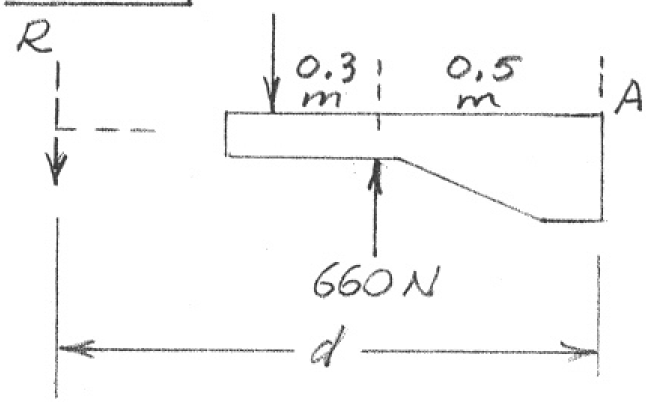

$$R = \Sigma F = 680 - 660 = 20 \text{ N}$$

$$Rd = \Sigma M_A$$

$$20d = 680(0.8) - 660(0.5)$$

$d = 10.70$ m to the left of A

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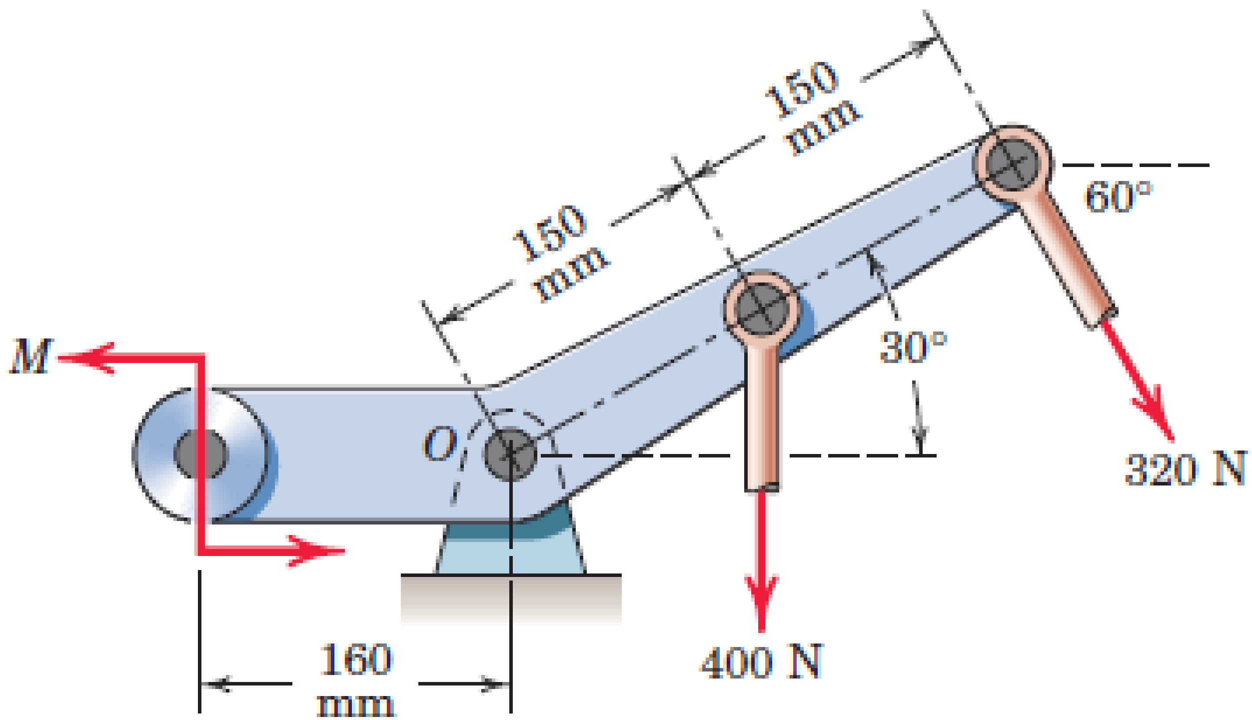
2/83 If the resultant of the two forces and couple M passes through point O, determine M.

Ans. $M = 148.0\ \text{N}\cdot\text{m CCW}$



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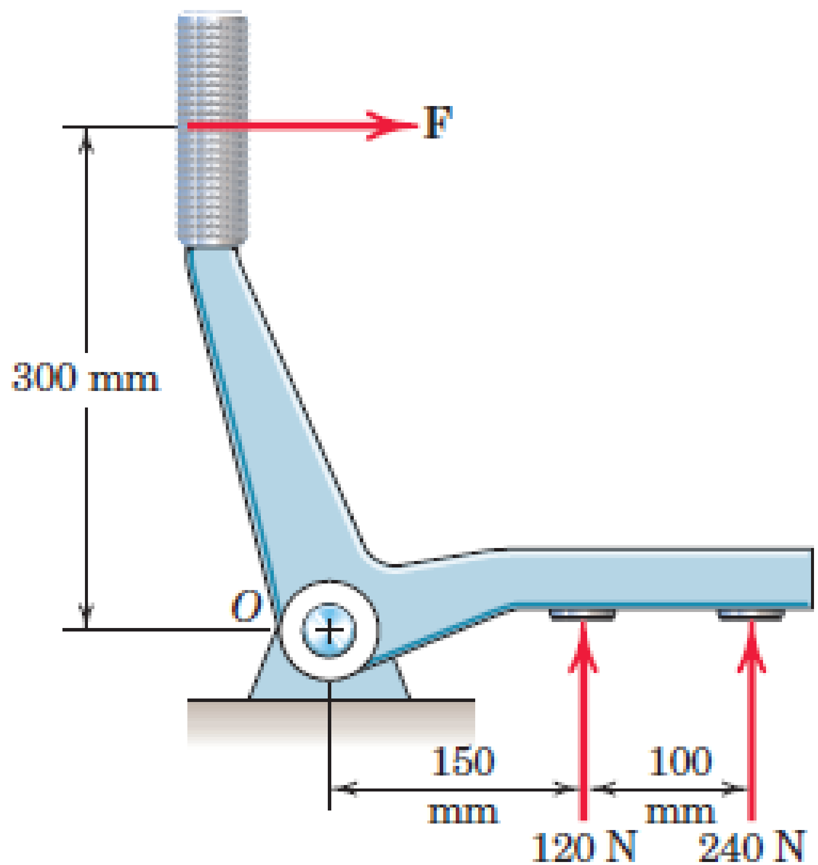
2/83 $M_O = 0$, so

$$\circlearrowleft^{+} M - 400(0.150 \cos 30^\circ) - 320(0.300) = 0$$

$$\underline{M = 148.0 \text{ N}\cdot\text{m}}$$

2/84 Determine the magnitude of the force **F** applied to the handle which will make the resultant of the three forces pass through *O*.



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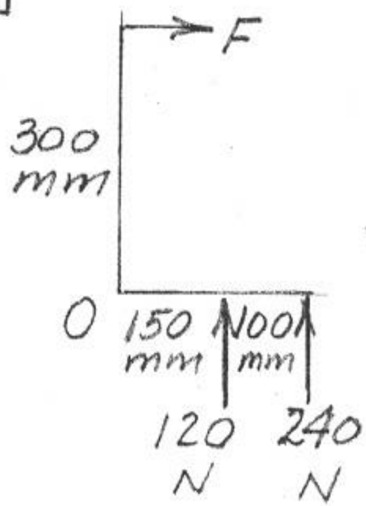

$\Sigma M_O = 0$ for R to pass through O

$$300F - 120(150) - 240(250) = 0$$

$$\underline{F = 260\ N}$$

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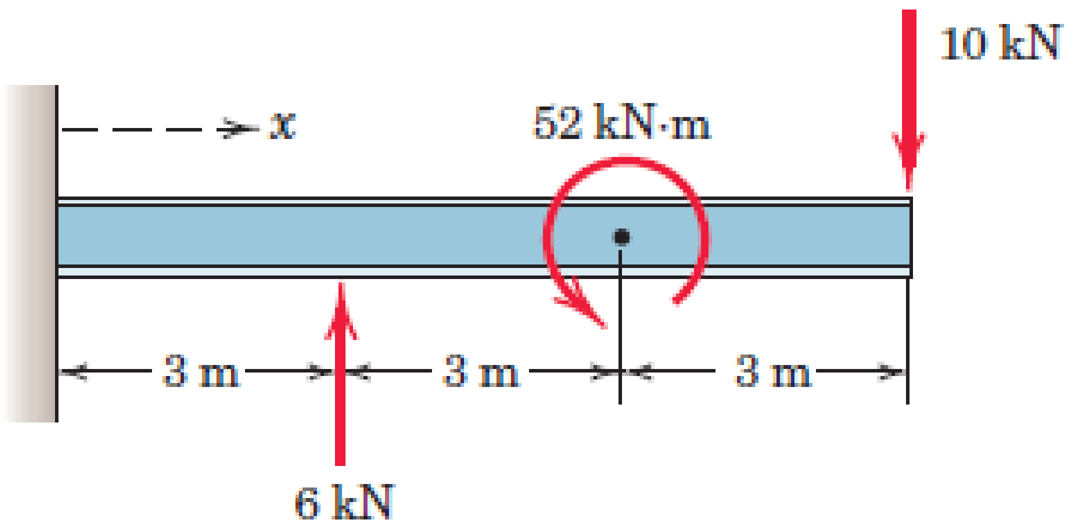
2/85 Determine and locate the resultant **R** of the two forces and one couple acting on the I-beam.

Ans. $R = 4$ kN down at $x = 5$ m



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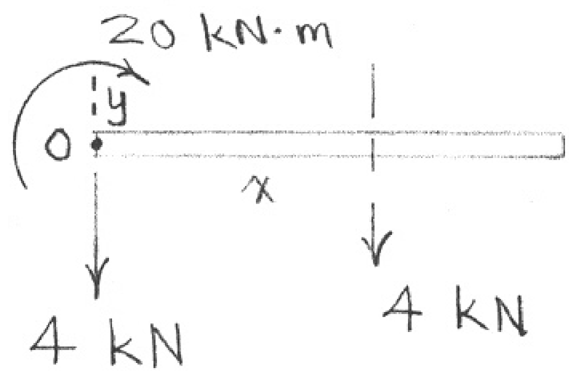
2/85 Force-Couple system at point O:

$$\underline{R} = \Sigma \underline{F} = (6-10)\,\underline{j} = -4\underline{j} \ \text{kN}$$

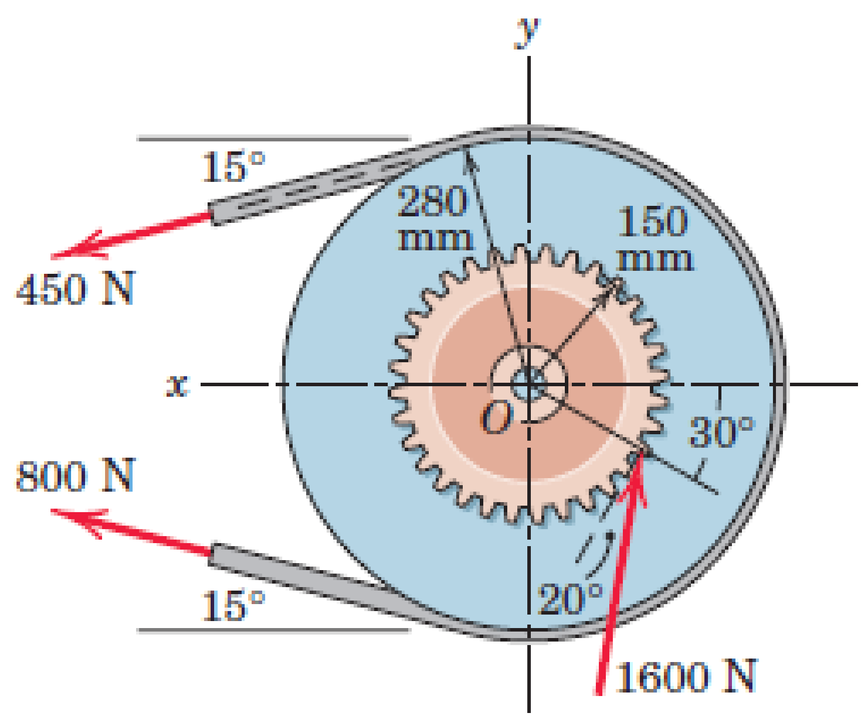
$$\circlearrowleft + \ M_O = 6(3) - 10(9) + 52 = -20 \ \text{kN}\cdot\text{m}$$



$$x = \frac{M_O}{R} = \frac{20}{4}$$

$$= \underline{5 \ \text{m}} \quad (\text{on beam!})$$

2/90 The gear and attached V-belt pulley are turning counterclockwise and are subjected to the tooth load of 1600 N and the 800-N and 450-N tensions in the V-belt. Represent the action of these three forces by a resultant force **R** at *O* and a couple of magnitude *M*. Is the unit slowing down or speeding up?



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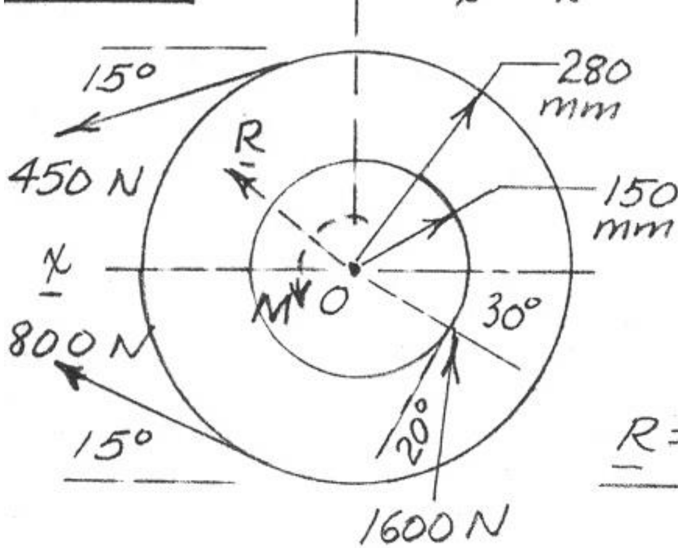
$R_x = \Sigma F_x = (800 + 450)\cos 15^\circ - 1600 \sin(30^\circ - 20^\circ)$

$= 1207 - 278 = 929.6\ N$



$R_y = \Sigma F_y = 1600 \cos 10^\circ$

$+ (800 - 450)\sin 15^\circ$

$= 1576 + 90.6 = 1666\ N$

$\underline{\mathbf{R} = 930\mathbf{i} + 1666\mathbf{j}\ N}$

$M = \Sigma M_O \ (\overset{\curvearrowleft}{+})$; $M = 1600 \cos 20^\circ(0.150) + (450 - 800)0.280$

$= 225.5 - 98.0 = \underline{127.5\ N\cdot m\ \ CCW}$

so unit is speeding up in CCW dir.

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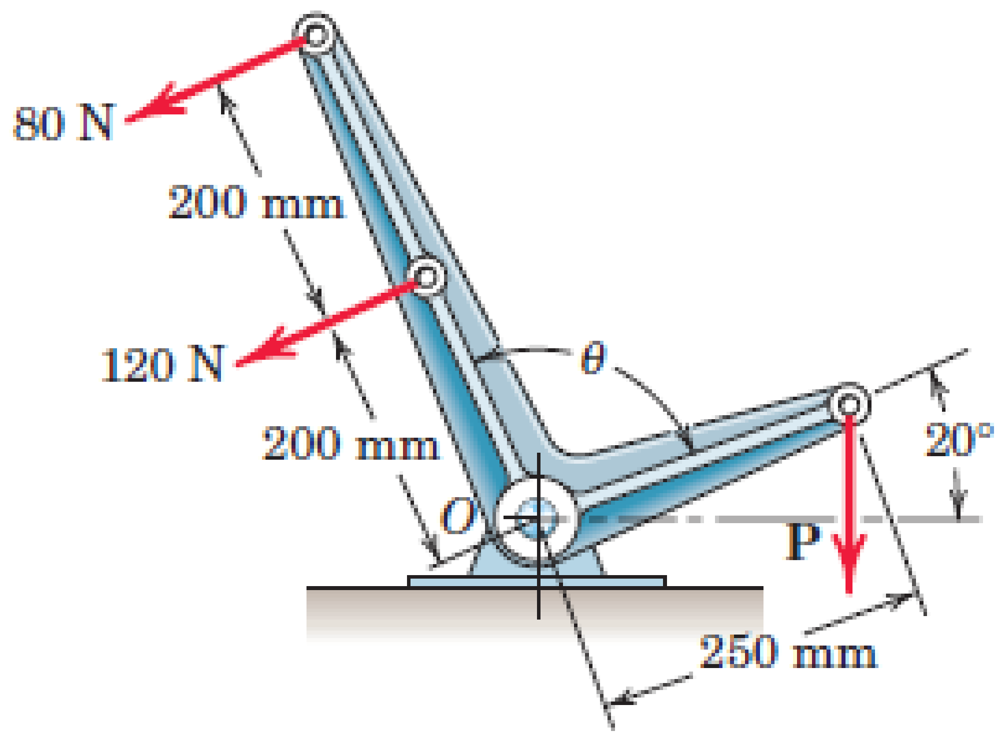
2/92 In the equilibrium position shown, the resultant of the three forces acting on the bell crank passes through the bearing *O*. Determine the vertical force **P**. Does the result depend on θ?



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2/92 $\Sigma M_O = 0$ since R passes through O

$$80(400) + 120(200) - 250 P \cos 20° = 0, \quad \underline{P = 238 \text{ N}}$$

Moment of 80- and 120-N forces unaffected by θ, so result for P is independent of θ.

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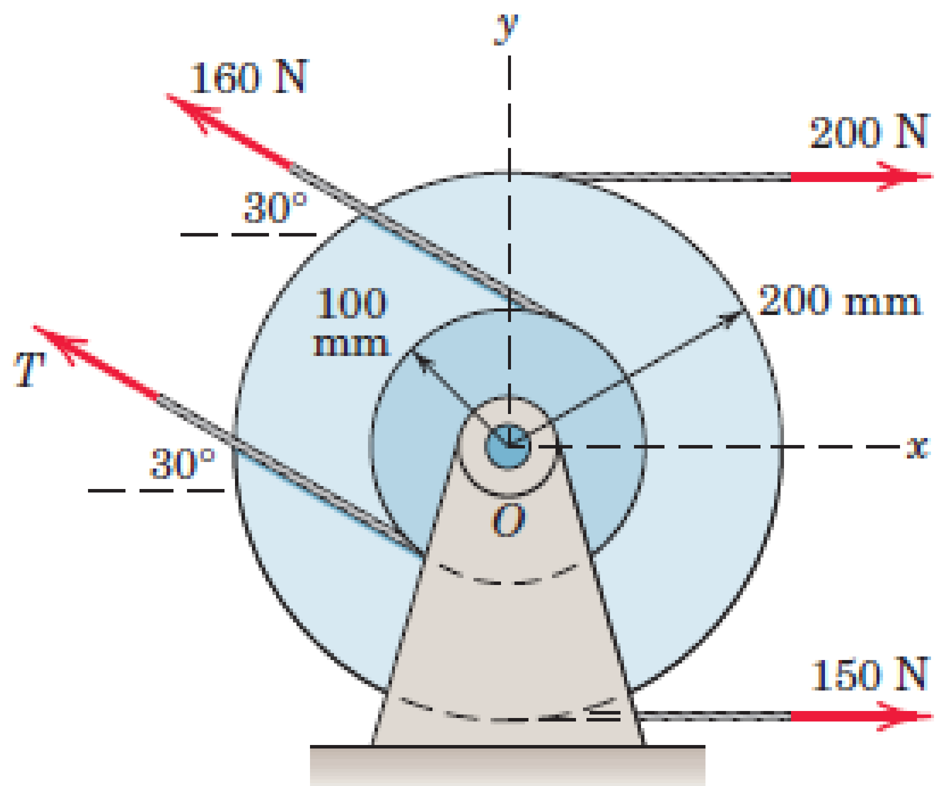
2/93 Two integral pulleys are subjected to the belt tensions shown. If the resultant **R** of these forces passes through the center O, determine T and the magnitude of **R** and the counterclockwise angle θ it makes with the x-axis.

Ans. $T = 60$ N, $R = 193.7$ N, $\theta = 34.6°$



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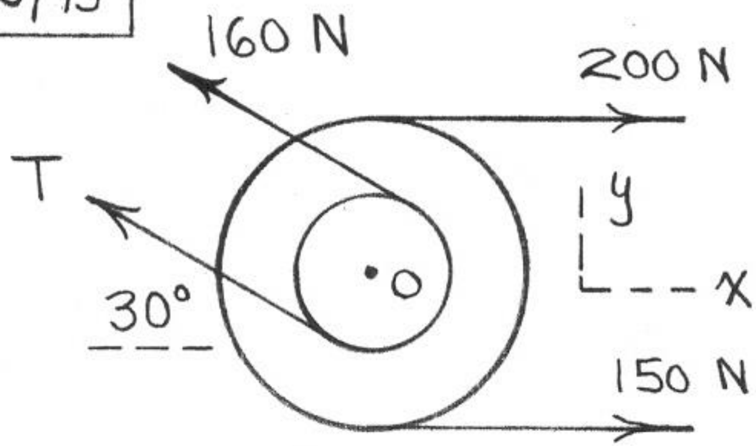
$$\begin{cases} r_1 = 0.1 \text{ m} \\ r_2 = 0.2 \text{ m} \end{cases}$$



$$+\circlearrowleft \Sigma M_O = 0 : \; 200(0.2) - 150(0.2) - 160(0.1) + (0.1)T = 0$$

$$\underline{T = 60 \text{ N}}$$

$$R_x = \Sigma F_x = 200 + 150 - (160 + 60)\cos 30^\circ = 159.5 \text{ N}$$

$$R_y = \Sigma F_y = (160 + 30)\sin 30^\circ = 110 \text{ N}$$

$$R = \sqrt{R_x^2 + R_y^2} = \underline{193.7 \text{ N}}$$

$$\theta = \tan^{-1}(R_y / R_x) = \underline{34.6^\circ}$$

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